Does Modelling Driver Adaptation Change Charging Infrastructure Decisions? A Comparison of Data Replay and Agent-Based Simulation for Minibus-Taxi Electrification in Cape Town

J. T. Sello[a], M. J. Booysen[a,b,c,*]

[a] *Electrical and Electronic Engineering, Stellenbosch University, South Africa*
[b] *Industrial Engineering, Stellenbosch University, South Africa*
[c] *Department of Computing and Technology, Uganda Christian University, Uganda*

Abstract

Electrifying minibus taxis requires charging infrastructure that accommodates both vehicle energy requirements and the operational decisions of drivers. An existing planning approach uses data-replay models to assess whether observed mobility is compatible with electrification. However, it remains unclear whether a fleet simulation model in which drivers' decision-making and agency are explicit suggests a different charging infrastructure configuration. This study compares a data-replay model with a decision-adaptive agent-based model for three minibus-taxi communities in Cape Town, South Africa, selected from eight communities identified using vehicle GPS telemetry. Both models electrify the same observed vehicles. The agent-based model also simulates the internal combustion engine (ICE) taxis on the same routes, which can carry the passengers that an electric vehicle (EV) misses while it charges. Each model selects a charging design with its own measures of operational performance, and each selected design is then evaluated in the other model.

Under a common reference design selected by the data-replay model, 57.0%, 22.6% and 49.0% of vehicle-days end in stranding in Khayelitsha, Bellville and Wynberg, respectively. In the base case, no agent-based run strands, because drivers can travel to charging stations. However, under the reference design, the EV agents carry 18.1%, 6.9% and 26.4% fewer passengers than in their ICE counterfactuals. When each model selects its own design, the agent-based model increases installed charging power from 250 to 650 kW in Khayelitsha, and keeps the same installed power but moves the connectors to other hubs in Bellville and Wynberg. When evaluated in the agent-based model, the data-replay designs carry 11.9, 3.5 and 10.3 percentage points fewer EV passengers than the agent-based designs. Where the share can be measured, ICE competitors take 85% to 98% of the passengers that the EVs lose. The differences persist in five of the six sensitivity cases in Bellville and in all cases in the other communities. In Khayelitsha, the additional capacity pays for itself at a fare of about R14 a trip; in Bellville and Wynberg, the agent-based design costs less and carries more passengers.

The results show that replay-based feasibility assessments do not necessarily identify the charging design that best supports passenger service when drivers can adapt. Agent-based modelling captures this additional operational response, but requires about three orders of magnitude more computation and explicit assumptions about driver behaviour and assumptions or information about passenger demand. The appropriate modelling approach therefore depends on the planning question and the operational constraints of the proposed electrification pilot.

Keywords: paratransit, minibus taxi, electrification, agent-based modelling, data replay, charging infrastructure, vehicle communities, mixed fleet, multi-objective design evaluation, Cape Town

*Corresponding author
Email addresses: jtsello@sun.ac.za (J. T. Sello), mjbooysen@sun.ac.za (M. J. Booysen)

1. Introduction

1.1. Paratransit electrification as a coupled planning problem

Paratransit systems, such as minibus taxis, are a critical component of urban mobility in many cities, particularly in developing countries where large gaps remain in formal public transport provision [1, 2]. As global efforts to reduce greenhouse gas emissions intensify, the electrification of these systems has become a focal point for sustainable urban transport planning [3, 4, 5]. However, the transition from internal combustion engine (ICE) vehicles to electric vehicles (EVs) introduces a complex set of interdependencies that were previously weakly coupled. Specifically, the mobility of vehicles is now closely linked to vehicle drive-train efficiency, battery capacity, and charging infrastructure, which in turn affects the electrical grid and drivers' ability to service passenger demand [6]. This coupling creates new challenges for planning and operation, as the limitations of EVs, such as slower charging times (440 km/min for diesel compared to 1.2–3.9 km/min for electric minibuses [6, 7]) and limited range (840 km for diesel [6] compared to about 130 km for the battery window of the electric minibus of this study, Section 4.1.1), naturally change the operational status quo, introducing opportunity costs that were not present for internal combustion engine (ICE) vehicles. When investigating the electrification of paratransit systems, it is therefore essential that the planning models account for the interactions between these systems, as well as the adaptive behaviour of drivers who respond to these constraints in their operational decisions [8, 9, 10].

In South Africa, minibus-taxi drivers have significant agency over their operational decisions, including where, when, and how long to operate [1, 11, 12, 13]. This agency allows drivers to adapt their mobility in response to changing passenger demand, traffic conditions, and other operational constraints [13, 14]. With the introduction of electrification, drivers must also learn to adapt their behaviour in response to the constraints imposed by electrification, such as limited range and charging availability [6]. As a result, the success of electrification initiatives is not solely dependent on the technical feasibility of EVs but also on how drivers respond to these new operational constraints. Understanding and modelling driver behaviour is therefore crucial for accurately predicting the outcomes of electrification and for designing effective charging infrastructure that meets both operational and passenger demand requirements.

Furthermore, because of significant economic, infrastructural, and operational barriers, mixed fleets are likely to be the norm in the early stages of paratransit electrification [15], where the first electric minibus taxis will be a small group of vehicles that share stops, routes, and hubs, rather than an entire electrified fleet. This means that, for the purposes of this study, the planning and evaluation of electrification initiatives are conducted at the community scale, where the interactions between electric and internal combustion engine (ICE) vehicles are particularly relevant. In such a mixed-fleet environment, the time spent by EVs at charging stations represents an opportunity cost, as ICE competitors can continue to carry passengers during these periods. Furthermore, vehicle communities are not necessarily aligned with taxi associations or licensed routes, and the study does not assign any association to a vehicle or community. Therefore, the comparison of fixed-mobility data-replay models and adaptive-mobility agent-based models is most relevant at the scale and in the market in which electrification pilots are likely to occur. Thus, understanding the implications of modelling paradigms and driver agency with respect to the interactions between electric and ICE vehicles is crucial for designing effective charging infrastructure and ensuring the success of electrification initiatives in paratransit systems.

1.2. Modelling paratransit: the tale of two paradigms

Modelling the electrification of paratransit systems is a complex problem that requires careful consideration of both technical and behavioural factors. In the domain of assessing the integration of EVs into paratransit systems, two primary modelling paradigms have emerged: fixed-mobility data-replay models and adaptive-mobility agent-based models. Each paradigm offers distinct advantages and limitations, and the choice between them can significantly impact the predicted outcomes of electrification initiatives.

Planning models are normative tools that shape transitions and encode assumptions about which outcomes planners can observe and which they cannot [8, 16]. The fixed-mobility data-replay paradigm is strongly grounded in observed behaviour, providing a direct assessment of whether the existing mobility patterns are compatible with electrification. However, it assumes that mobility will largely persist after electrification, potentially overlooking the adaptive responses of drivers to new constraints. In contrast, the adaptive-mobility agent-based paradigm can represent these adaptive responses, allowing for a more dynamic

assessment of how drivers may alter their operations in response to electrification, as well as evaluating well defined counterfactual scenarios. However, this approach introduces additional behavioural assumptions, calibration requirements, computational complexity, and uncertainty into the modelling process. As a result, the choice between these two paradigms can significantly influence the predicted outcomes of electrification initiatives, particularly in terms of charging infrastructure requirements and passenger service levels.

1.3. Research gap

This research investigates the implications of driver agency for the planning of paratransit electrification pilots. Existing techniques can evaluate whether observed paratransit mobility is compatible with electrification, while behaviourally explicit models can represent adaptive responses to energy and charging constraints. What remains unclear is whether these different representations lead to materially different infrastructure planning decisions when applied to the same empirically observed vehicles, and whether a charging design suggested by a fixed-mobility model remains effective once drivers respond endogenously and compete for passengers with the internal-combustion vehicles that share their routes. It is also unclear how large such differences are, and what form they take, across the distinct operating communities in which electrification pilots are likely to take place.

1.4. Research questions

This study poses and addresses the following research questions:

1. For the same observed community vehicles, how do predicted electrification outcomes differ when mobility is represented as fixed empirical activity (M_R) versus endogenous driver behaviour (M_A) under a common reference charging design?
2. Does a model in which drivers' decisions are explicit suggest a different charging infrastructure configuration from a data-replay model, when each model judges designs by the outcomes it can validly represent?
3. What does a replay-derived charging design cost in passenger service when it is evaluated in the agent-based model, and what share of passengers do the electric vehicles lose to internal-combustion competitors while they charge?
4. How large is the difference between the models in three different vehicle communities, what form does it take (more capacity, connectors at other hubs, or both), and how sensitive are the size and the form to the number of drivers on the same routes and the vehicle assumptions?

1.5. Contributions

This study examines the implications of explicit behavioural representations of driver agency for paratransit electrification planning. It contributes to the literature in five ways. First, it introduces a comparison framework in which each model selects a charging design in its own model-native objective space, and each selected design is then evaluated in the other model, rather than forcing a single objective space onto models with different behavioural assumptions. Second, it applies this framework as a controlled comparison of fixed-mobility data replay and decision-adaptive agent-based modelling, in which the same observed vehicles are electrified in both models, in three minibus-taxi communities in Cape Town. Third, it introduces the vehicle community, found from router GPS data and screened for electrification by a rule that is frozen before any agent-based run, as a reproducible unit for the selection of electrification pilots. Fourth, it simulates every driver on the routes of the electrified vehicles in the agent-based model, with EVs in a mixed fleet, which makes the passengers that EVs lose to internal-combustion competitors measurable. Fifth, it enumerates every candidate design in the replay model, so the replay Pareto front is exact, and it validates the agent-based candidates on paired seeds and evaluates the selected designs on seeds that the selection did not use.

2. Related work

2.1. Electrification feasibility under observed mobility

Assessing the feasibility of electrifying a fleet of vehicles requires an understanding of the energy requirements of the vehicles and the availability of charging infrastructure. The most common way to obtain both is to replay the observed movement of the fleet through an EV energy model. GPS logs and travel surveys

of private vehicles show that most daily driving fits within the range of an affordable EV that charges once a day [17], and the dwell patterns of taxi fleets have been used to site and size charging stations [18, 19, 20]. In each case the trips are fixed, and a vehicle can charge only where it already stops.

For minibus taxis in sub-Saharan Africa, data replay has become a significant planning method, partly because vehicle tracking is the only data that records the duty cycle on which electrification depends [21]. Tracked vehicles have been used to estimate energy consumption [22, 23, 24, 7], to assess charging opportunities with grid and solar supply [25, 26, 27], and to measure the share of vehicle-days that remain viable under different charging designs [28, 29, 30], which can differ sharply between groups of drivers in the same town [31]. In Cape Town, replay underpins estimates of energy demand, emissions and operator costs [27]. More recently, trip GPS trajectories have been used to empirically measure the energy consumption of the first commercial electric minibus taxi in South Africa [7].

Underlying these studies is the assumption that mobility persists after electrification. A vehicle-day that cannot be completed on battery power is written off rather than adapted, so the conclusion that a mixed fleet is needed follows partly from the assumption itself [8]. Scheduled-mobility models relax this assumption by optimising vehicle schedules and charging centrally, including the joint deployment of electric and ICE vehicles [15, 32], but they describe a reformed system rather than current operation [6]. Neither paradigm represents drivers who compete for passengers, so neither can measure the passengers that an EV loses to ICE competitors while it charges. Data replay therefore answers whether today's trips can be electrified, not how operation would change once they are.

2.2. Behavioural simulation of taxi and paratransit fleets with charging

Agent-based and discrete-event simulations relax the persistence assumption by letting vehicles decide when and where to charge. Agent-based transport simulations represent electric vehicles whose drivers plan charging around their activities, such as the electric vehicle extension of MATSim [9, 33] and BEAM, which needed a detailed model of the charging decision to reproduce observed charging [10]; agent-based simulation has also been used to site public chargers [34]. Simulations of electric taxi and ridesourcing fleets show that EVs can match the service of ICE fleets given adequate charging infrastructure [35, 36], and simulations of autonomous fleets have been used to jointly size the fleet and its charging facilities [37, 38]. Observed charging supports this focus on behaviour: in Shenzhen's fully electric taxi fleet, drivers organise their charging around their shifts and concentrate on a few stations [39]. How this behaviour is represented also changes the infrastructure that is planned. In an agent-based simulation of Sweden, realistic charging archetypes required 2.3 to 4.5 times more public chargers per vehicle than refuelling-style charging [40], yet supply-side planning studies still rely mostly on simplistic behavioural assumptions [41, 42].

Paratransit simulation has a shorter history. Co-evolutionary operator models in MATSim reproduce the main minibus corridors of a South African metropolitan area [43], agent-based simulation shows that paratransit operators gain most from long headways in formal transit [44], and Kampala's minibus taxis have been modelled as a complex adaptive system [45]. These models are consistent with evidence that drivers' working hours respond partly to daily revenue [13] and that drivers search widely for passengers [14], but none of them represents electric vehicles. To our knowledge, only one framework simulates electric minibus taxis whose drivers choose their routes, waiting and charging in response to remembered demand and battery state [46], and this study builds on it.

The price of behavioural realism is calibration. An agent-based model must reproduce several observed patterns at once [47], and there is no generally accepted procedure for validating it [48], a burden that replay avoids by construction. The formal-fleet models also simulate either a fully electric fleet or a single operator, so they cannot measure the passengers that an EV loses to ICE competitors while it charges. In paratransit, mixed fleets have so far been treated as a scheduling problem rather than as competition [15, 32]. Yet early pilots will place a few EVs among many ICE vehicles on the same routes, where this competition determines what electrification costs the drivers who adopt it first.

2.3. Paratransit networks and operating communities from GPS data

Where paratransit has no formal route data, GPS and phone traces have been used to reconstruct it. Mobile-phone mapping produced open network data for Nairobi's matatus [49], and collaborative mapping in Nairobi and Maputo has made minibus systems visible in planning [50]. In South Africa, more than 500 Cape Town routes have been clustered into trunk, intermediate and feeder routes [51], and GPS traces

from Tshwane have been extrapolated to minibus-taxi trip counts per street segment [52], while traces from Kampala reveal how drivers search for passengers [14]. This work builds on general methods for stop detection and trajectory mining [53], and its units of analysis are routes, corridors and road segments. Vehicles have been grouped by home location [31], but not by the hubs and routes that they share, which is the unit at which a charging pilot would be planned.

2.4. Comparing models of different structure

Comparing models of different structure is an established practice in agent-based modelling. Docking tests whether two models of the same system produce equivalent results, at the level of numerical, distributional or relational equivalence [54]. Replication studies show that independently built models of the same system can disagree for reasons of implementation, and they define criteria for when two models match [55, 56]. Independent replication can expose artefacts that a single implementation hides [48]. In minibus-taxi electrification, the alignment of two energy-simulation tools reconciled consumption estimates that differed by a factor of more than two [57]. These comparisons ask whether models agree on a common output. Data replay and agent-based simulation do not share one: replay reports whether a vehicle-day strands, whereas the agent-based model, for this study, focuses on how many passengers an EV carries. The choice of model therefore shapes which outcomes planners can see [16, 6, 8]. A fair comparison lets each model select a design by its own measures and then evaluates both designs in the model that can represent the outcomes of both. Only the agent-based model includes ICE competitors, the ICE taxis that share the EVs' routes; Section 4.1 explains why and how the designs are still compared like-for-like.

Table 1 positions the studies discussed against this one. Replay studies of minibus taxis observe the vehicles that would be electrified but hold their movement fixed. Behavioural models let drivers adapt, but they have not been compared with replay on the same vehicles. Scheduling studies place EVs in a mixed fleet without competition, and only Sello and Booysen [46] has simulated EVs that compete with ICE vehicles for passengers. No study has electrified the same observed vehicles in both paradigms, selected charging infrastructure in each by its own measures, and evaluated the resulting designs where EVs compete with ICE vehicles for passengers. This study addresses that gap through the research questions in Section 1.4.

Table 1: Representative studies of fleet electrification by the way they represent mobility and charging. Replay: observed trajectories; scheduled: centrally optimised operation; adaptive: driver or dispatcher decisions.

Study	**Fleet**	**Mobility**	**Charging infrastructure**	**Passenger service**	**ICE competition**	**Same vehicles in both paradigms**
Pearre et al. [17]	Private cars	Replay	Not designed	No	No	No
Tu et al. [19]; Yang et al. [20]	E-taxi	Replay	Optimised	No	No	No
Bischoff and Maciejewski [35]; Bauer et al. [36]	E-taxi, ridesourcing	Adaptive	Evaluated, sized	Yes	No	No
Chen et al. [37]; Yang et al. [38]	Autonomous fleet	Adaptive (dispatch)	Optimised	Yes	No	No
Liao et al. [40]	Private cars	Adaptive	Sized	No	No	No
Giliomee et al. [23]; Pretorius et al. [29]	Minibus taxi	Replay	Evaluated	No	No	No
Pretorius et al. [15]; Wust et al. [32]	Minibus taxi	Scheduled	Evaluated	No	Mixed fleet, no competition	No
Sello and Booysen [46]	Minibus taxi	Adaptive	Evaluated	Yes	Yes	No
This study	Minibus taxi	Replay and adaptive	Selected in each, cross-evaluated	Yes (M_A)	Yes (M_A)	Yes

3. Study area and data

3.1. Minibus-taxi operation in Cape Town

Minibus taxis carry the largest share of the public-transport trips in Cape Town [58, 59]. Their operation is shaped by vehicle owners, taxi associations and the provincial operating-licence system [1, 12]. Five operating

characteristics are particularly relevant to electrification and to the models developed in this study.

First, passengers commonly board at taxi ranks, where vehicles queue on a first-in, first-out basis and depart when sufficiently full [1]. This is known as fill-and-go loading. Second, passengers may board or alight at stops along a route, although boarding is generally concentrated at terminals. Third, vehicles tend to operate on a limited set of routes between established terminals [60, 51]. Fourth, drivers may stop and wait for more passengers, or move slowly along a route to find them, a practice referred to here as hold-back [14]. Finally, demand is directional and varies over the day. During the morning peak, many passengers travel from residential areas towards employment centres, leaving some taxis with relatively few passengers on the return journey [59]. The pattern often reverses in the evening.

These features create specific charging constraints. Long dwell periods at ranks and terminals provide potential charging opportunities, but a taxi that leaves a loading queue to charge may lose its position and the passengers it would otherwise have carried. The models therefore need to represent both the availability of charging opportunities and the cost of using them.

Shown in Figure 1a, the study area is the union of the City of Cape Town metropolitan municipality and the surrounding Stellenbosch and Drakenstein municipalities. Telemetry data outside the area is flagged but not deleted. Around 19% of the driven distance lies outside the study area, and a vehicle-day with 50 km or more outside the area is considered a long-distance day.

3.2. Vehicle telemetry

The observed fleet dataset is a subset of the Sebenza router telemetry [63], which is a GPS dataset of minibus taxis in South Africa. The subset consisted of 378 minibus taxis operating in Cape Town over 11 weekdays from 1 to 15 December 2025, resulting in a total of 3,916 vehicle-days. The festive period from 16 December was excluded, because it produces a distinct mobility regime [64]. Each GPS point includes a pseudonymised vehicle identifier, a timestamp, a position, and a quality flag. The median sampling interval is 79 seconds, with the 5th to 95th percentile ranging from 60 to 81 seconds. On average, a moving taxi covers a median distance of 435 meters between two points. While the start, end, and overnight positions of each day are observed, stops shorter than 2 to 3 minutes, which include most boarding and alighting events along a route, are not captured in the dataset.

Table 2 summarises the duty cycle of the observed Cape Town fleet. The median valid vehicle-day covers 190 km (interquartile range 130 to 247 km), whereas the battery window of the study EV gives about 130 km; only 24.9% of the valid days fall within this range. The median driving window, from the departure after the first stop of the day to the arrival at the last, is 17.4 h. The median taxi therefore parks for about 6.6 h, in which a 3.7 kW socket adds at most about 22 kWh, less than a third of the median daily need of 72 kWh. Further analysis shows that long dwells concentrate at ranks, official route termini, and stations. Additionally, each vehicle varies strongly from day to day: the intraclass correlation is 0.43 (95% confidence interval 0.36 to 0.50) for the daily distance and 0.62 (0.56 to 0.67) for the daily moving time, thus 57% of the variance of the daily distance lies between the days of one vehicle.

Table 2: Duty cycle of the Cape Town fleet by day class. Term weekdays: 1–5 and 8–10 December 2025; school-holiday weekdays: 11, 12 and 15 December. Long-distance: the share of all vehicle-days with 50 km or more outside the study area. Distance and driving window: over the valid vehicle-days, with the interquartile range in brackets; the driving window runs from the end of the stop that contains the first point of the day to the start of the stop that contains the last point. Within range: the share of the valid vehicle-days of 130 km or less, the distance of the EV battery window.

Day class	Vehicles	Vehicle-days	Valid (%)	Long-dist. (%)	Distance (km) Median [IQR]	Driving window (h) Median [IQR]	Within range (%)
Term weekdays	374	2,856	61.8	2.5	189 [128–242]	17.5 [15.2–19.9]	25.8
School-holiday weekdays	367	1,060	64.2	7.5	196 [136–259]	17.3 [15.2–19.5]	22.6
All study weekdays	378	3,916	62.5	3.9	190 [130–247]	17.4 [15.2–19.8]	24.9

To maintain privacy, the study uses only pseudonymised data. This paper reports aggregates only and never assigns a taxi association to a vehicle or a community. Additionally, home locations are neither named nor shown on maps.

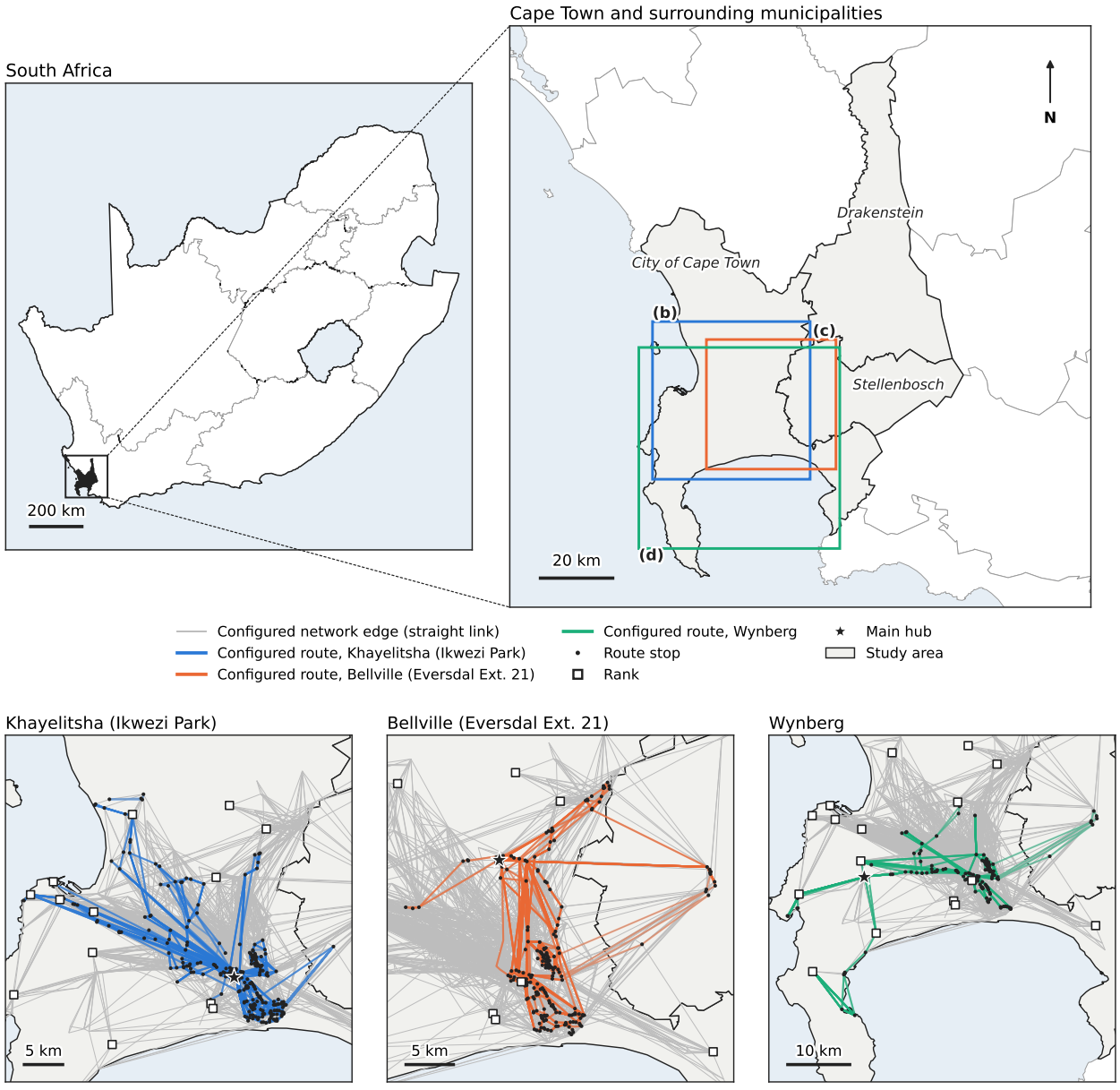


Figure 1: Study context. (a) South Africa with the study area and a zoom on Cape Town and the surrounding municipalities (Census 2011 boundaries [61, 62]); the coloured boxes give the extent of (b) to (d). (b) Khayelitsha, (c) Bellville and (d) Wynberg: the configured network of each community as straight links between nodes (grey), the configured routes of the community drivers as straight links between their stops (colour), the route stops (dots), the formal ranks (squares) and the main hub (star). Home nodes and the links to them are not shown.

3.3. Infrastructure, planning and survey data

Table 3 lists the public data that complement the telemetry. The formal stops combine the City of Cape Town route termini, and the taxi ranks and railway stations in OpenStreetMap. The 2023–2028 Comprehensive Integrated Transport Plan (CITP) of the City of Cape Town [59] provides the daily boardings at minibus-taxi facilities, which set the level of passenger demand, and the morning-peak vehicle trips by direction on surveyed corridors, which set its direction. The provincial route register [65] gives only an upper bound of the fleet size at a hub, because one licence can authorise many routes.

Table 3: Public data used with the telemetry.

Source	Content	Use
City of Cape Town (2021) [60]	Minibus-taxi route termini	Formal nodes
OpenStreetMap, July 2026 [66]	Taxi ranks and railway stations; drivable roads; land use	Formal nodes; edge lengths; residential ends of routes
Census 2011 [61, 62]	Municipalities, main places and subplaces	Study area; place names
CITP, Table 3-9 [59]	Daily boardings at minibus-taxi facilities	Scale factor k
CITP, Table 3-18 [59]	Morning-peak vehicle trips and passengers by direction on surveyed corridors	Load factors; demand audit
Provincial route register (2022) [65]	Valid operating licences by route	Upper-bound check of k
School calendar 2025 [67]	Last school day (10 December)	Day classes

3.4. Observation period, data quality and analysis sample

Established by Pretorius et al. [29], an operational vehicle-day runs from 04:00 to 03:59 the following morning. This boundary groups overnight charging with the operating cycle that precedes it, since most taxis are parked during this period. A GPS step is treated as evidence of data loss, and the vehicle-day is marked invalid, when consecutive observations are more than 600 seconds apart and more than 1,000 m apart. A long interval without corresponding movement is instead treated as a parked period.

Of the 3,916 observed vehicle-days, 2,446 (62.5%) satisfy the validity criteria. The valid share varies from 29% to 87% across dates, while the number of active vehicles remains comparatively stable at 351–360 per day. This pattern suggests that variation in the valid share reflects data completeness more than changes in fleet activity. The agent-based model therefore applies a day-level correction factor when scaling observed activity to daily operation.

The study days fall into two classes (Table 2). The eight term weekdays (1 to 5 and 8 to 10 December) end with the last school day, and the three school-holiday weekdays (11, 12 and 15 December) follow it [67]. The holiday weekdays have three times the share of long-distance days (7.5% against 2.5%) and a longer mean distance (214 against 193 km), whereas without the long-distance days the two classes are almost the same (mean 184 against 185 km). They therefore represent a different demand regime and are excluded from every fit; only the structural stages (stop events, network, communities and routes) use them, and the calibration and the replay use the term weekdays. Of the term weekdays, Tuesday 2 and Wednesday 10 December, with valid shares of 79% and 86%, are the hold-out days, and the other six are the calibration days. Section 4.1 gives the steps that use each class. No candidate community is excluded on data quality. Their valid shares are 59% to 75%, their gap times 1% to 2%, and their shares of driving outside the network 14% to 47%.

4. Methodology

4.1. Comparative design and shared assumptions

The study evaluates the same observed vehicles using two models, summarised in Table 4. In the data-replay model (M_R), each vehicle is replayed as an observed EV along its recorded movements. In the agent-based model (M_A), drivers generate realistic movements through operational decisions and can respond to energy and charging constraints.

In M_A, the EV agents compete for passengers with ICE competitors: synthetic agents that follow the same decision model and operate on the same routes (Section 4.5). M_R has no ICE competitors, for two reasons. First, replayed movement is fixed and passengers are not represented, so other vehicles could not change any outcome of M_R. Second, in M_A the competitors are what make charging costly: while an EV

charges, they can carry the passengers that it would have carried. The designs are still compared like-for-like, because $\mathbf{x}_R^\star$ and $\mathbf{x}_A^\star$ are both evaluated in M_A against the same ICE competitors (Section 4.7.3).

The representation of vehicle movement is the primary difference under investigation. Both models use the same vehicle and charging assumptions, candidate sites and design space, and both electrify the same vehicles in each community: its qualified vehicles, the observed vehicles that keep at least one eligible route of 5 km or more (Section 4.2.2). Where their operating periods differ, the evaluation windows are made as comparable as the model structures allow. One asymmetry remains: M_R allows vehicles to charge during recorded dwell periods that include waiting in a rank loading queue, whereas M_A requires drivers to leave the queue to charge. This assumption favours M_R and should be considered when interpreting the results.

Table 4: Elements of the two models. Movement is the treatment of the comparison; the other elements are the same in both models, made equivalent, or present in one model only.

Element	M_R	M_A	**Relation**
Electrified vehicles	The qualified vehicles of the community, replayed as observed EVs	The same vehicles, as EV agents	Same
Other vehicles	None; replayed movement does not interact	ICE competitors: synthetic copies of the EV agents	M_A only
Movement	Observed and fixed	Output of driver decisions	Treatment
Days	The eight observed term weekdays	One typical term weekday, simulated for 14 evaluation days	Equivalent
Vehicle and charging values	One assumption set	The same assumption set	Same
Candidate sites and designs	The first three hubs; (N_1, N_2, N_3, P)	The same	Same
Overnight charging	3.7 kW where the vehicle parks	3.7 kW at the home node	Equivalent
Charging in a rank loading queue	Yes	No	Asymmetry
Passenger service	Not represented	Requests, waits and the passengers of each driver	M_A only
Selection rule	At least 95% of the service of the best design, then the lowest peak site load and installed power	The same rule, with a passenger measure of service	Same rule

4.1.1. Vehicle and charging values

Both models use one assumption set based on the Higer H5C, a 15-seat electric minibus [68]. The vehicle has a usable battery capacity (C) of 70.47 kWh, an empirically measured energy consumption rate of 0.38 kWh/km in loaded operation [7], a vehicle charging limit of 60 kW and a charging efficiency of 90%. The battery operates between a reserve of $0.2\,C$ and a charge limit of $0.9\,C$. This window holds 49.3 kWh, or about 130 km. Every vehicle charges overnight at 3.7 kW from a socket outlet, and site chargers serve vehicles in a first-come, first-served queue.

4.1.2. Unit of analysis and study days

The unit of analysis is the vehicle community: a group of vehicles with overlapping operating areas, routes and hubs. These shared operating patterns create overlapping charging requirements and make communities a useful scale for evaluating electrification pilots. A community is a data-derived operating group, not a taxi association, fleet operator or licensed route.

Network construction, route inference and community detection use all 11 observation weekdays. Agent-based model calibration uses the six designated calibration days, while the two hold-out days remain unused until the calibrated model has been accepted. Since M_R has no fitted behavioural parameters, it replays all eight term weekdays.

4.1.3. Planning principle

The study selects one charging design per model and community. The selection principle is to achieve service performance close to the best observed within the model's feasible design set while minimising the required infrastructure. Installed power, calculated as the number of connectors multiplied by their rated

power, serves as a proxy for infrastructure cost during the selection; the selected designs are then priced with a cost model (Section 4.7.3).

The service measure depends on the model. In M_R, the selection rule uses the share of vehicle-days that do not strand under repeated operation, in the steady state described by Pretorius et al. [29]. It does not use their viable share, which also counts a day below the $0.2\,C$ reserve as a failure, because in M_A a level below the reserve is normal behaviour: drivers look for a charger at that level. Both models therefore select by the same failure, stranding, and M_R reports its non-viable share beside it. In M_A, it uses the passenger throughput of the EV agents, alongside constraints on expired passenger requests. The measures are defined below, and the precise selection rules are given in Section 4.7.2. Since the models represent different aspects of operation, the selected designs are subsequently evaluated in a common agent-based framework, and each selected design is also evaluated in the replay model.

4.1.4. Model-native measures

Each model evaluates designs using the measures it can represent, as summarised in Table 5. The data-replay model focuses on the share of vehicle-days that strand and the peak charging load across the candidate sites. The agent-based model reports these measures and also represents passenger demand, waiting times, expired requests and the passenger throughput of the EV agents. Because an agent-based run is too expensive to evaluate every design, M_A searches the design space with a tree-structured Parzen estimator (TPE) [69], a sequential optimiser that proposes each new design from the results of the earlier ones (Section 4.7.2).

The two models also record a low battery differently. In M_R, a vehicle-day whose steady-state battery level falls below the $0.2\,C$ reserve, or that strands, is non-viable [29]. In M_A, there is no steady state, and drivers look for a charger at the reserve; the low-battery share counts the EV agent-days on which a route run ends or a charging session starts below $0.2\,C$. It describes how close drivers run to the reserve and is not a failure.

Table 5: Model-native measures of a charging design. All measures use the evaluation days of M_A or the replayed term weekdays of M_R. Role of each measure: Pareto front, an objective of the exact Pareto front of M_R over all designs; search, an objective of the TPE search of M_A; selection, used by the selection rule (Section 4.7.2); reported, computed but not used to choose a design. A dash shows a measure that the model cannot represent.

Measure	**Definition**	M_R	M_A
Stranded share	EV vehicle-days whose battery level reaches 0	Pareto front and selection	Reported
Non-viable share	EV vehicle-days below $0.2\,C$ or stranded in the steady state	Reported	–
Low-battery share	EV agent-days with a route end or session start below $0.2\,C$	–	Reported
Peak site load (kW)	Peak of the summed grid power of all site sessions	Pareto front and selection	Search and selection
Site and home energy (kWh a day)	Grid energy at the sites and at the homes	Reported	Reported
Expired share	Expired requests over all requests (EVs and ICE vehicles)	–	Search and selection
Mean passenger wait (min)	Time from the request to the boarding of served requests	–	Search
EV passengers carried	Passengers that the EV agents carry	–	Search
EV passenger change	EV passengers against the EV agents in their ICE counterfactual	–	Selection
EV passenger loss ℓ	EV passenger change corrected for the time away from home	–	Reported

In the replay model, the stranded share is the fraction of conclusive vehicle-days that end in stranding. The peak site load is the maximum simultaneous charging load summed across all sites. These measures define the replay model's design-selection objective space. In the agent-based experiments, every EV run is paired with an ICE counterfactual using the same configuration, random seed and warm start (Section 4.4). In this ICE counterfactual, the EV agents operate as ICE vehicles. The paired runs use the same seed, which reduces the simulation noise in the comparison.

The EV passenger change and the EV passenger loss are

$$\Delta_{\mathrm{EV}} = \frac{M_{\mathrm{EV}}^{\mathrm{EV\ run}}}{M_{\mathrm{EV}}^{\mathrm{ICE\ run}}} - 1, \qquad \ell = 1 - \frac{R^{\mathrm{EV\ run}}}{R^{\mathrm{ICE\ run}}}, \qquad R = \frac{M_{\mathrm{EV}}/M}{Y_{\mathrm{EV}}/Y}, \tag{1}$$

where M is the number of passengers carried by all drivers on the evaluation days, Y is the time that all drivers spend away from home, and the subscript EV denotes the EV agents. ℓ corrects Δ_{EV} for a change in the EV agents' time on the road. It is therefore a measure of passenger throughput relative to operating time, not a measure of passengers lost by the transport system as a whole.

4.1.5. Pipeline

Figure 2 summarises the study's methodology. GPS telemetry is used to infer the operational network, construct a route library and identify vehicle communities. A predefined screening rule selects three communities for further analysis (Section 4.2.3). For each community, the same qualified vehicles are electrified in both models.

The data-replay model evaluates every candidate charging design and selects a reference design, $\mathbf{x}_R^\star$. The agent-based model searches the same design space using its own passenger-service and infrastructure measures, producing a selected design $\mathbf{x}_A^\star$. Both designs are then evaluated in the agent-based model on seeds that the selection did not use, and both are also evaluated in the replay model. This cross-evaluation measures the operational consequences of selecting infrastructure under fixed-mobility assumptions. The analysis also examines whether these consequences persist across the three communities and under alternative vehicle energy-consumption, charging-policy and fleet-scale assumptions.

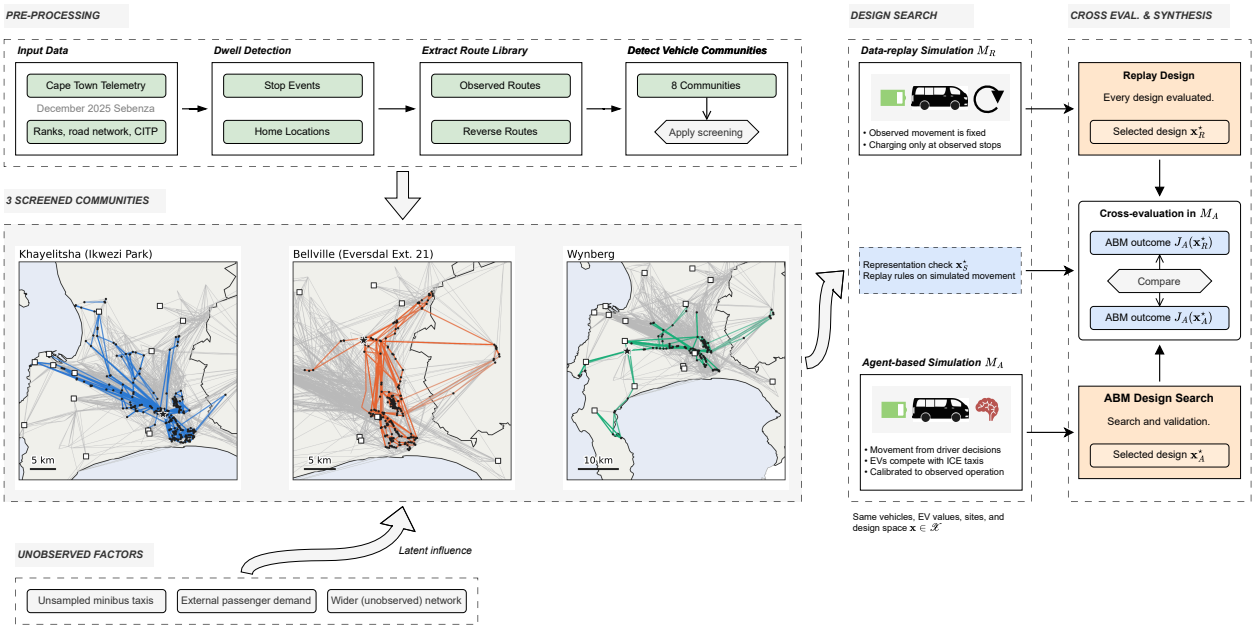


Figure 2: Overview of the study design. GPS telemetry is used to reconstruct the operational network, infer routes and identify vehicle communities. Three communities are selected using a predefined screening rule. The same qualified vehicles are electrified in the data-replay model (M_R) and the agent-based model (M_A); the latter also simulates ICE competitors on the same routes. The data-replay model evaluates all candidate charging designs and selects $\mathbf{x}_R^\star$. The agent-based model searches the same design space and selects $\mathbf{x}_A^\star$. Both designs are evaluated in the agent-based model, and their passenger-service outcomes are compared. The representation check $\mathbf{x}_S^\star$ applies the replay rules to the simulated movement of M_A. Both models share the vehicles, EV values, candidate sites and design space $\mathbf{x} \in \mathcal{X}$. The three communities are then compared to assess the size and the form of the differences.

4.2. Operational network, routes and vehicle communities

4.2.1. Stop events and the operational network

Stop events are extracted from the GPS trajectories with an anchored-dwell algorithm that tolerates isolated positioning errors. An event has at least three inlier points within 30 m of its anchor and lasts

at least 180 s. At the sampling interval of about 80 s, stops shorter than 2 to 3 min, which include most boarding and alighting along a route, are not observable. The home of a vehicle is the location within 300 m of which the first and last points of at least 75% of its valid vehicle-days lie.

One operational network is inferred and applied to all communities. Its nodes come from evidence of stopping where grid cells with stop events from at least five vehicle-days and three vehicles, merged by complete linkage within 200 m, together with the formal ranks, termini and stations that the fleet uses. Its directed edges come from evidence of driving which constitute hops between nodes on at least two vehicle-days, their reverse edges, and links between nodes within 300 m. Edge lengths are shortest drivable paths on OpenStreetMap. After one manual review of 211 flagged items, the network has 726 nodes, of which 18 are ranks. It captures 76% of the stop events and represents 66.5% of the distance driven between stops with respect to the full fleet.

4.2.2. Route library and driver route sets

Typically, minibus taxis load at a terminal and depart when full [1]. A trip therefore runs between two detected anchor dwells, consisting of stop events of 10 min or more at non-home nodes. The trips of each terminal pair are grouped by the Jaccard similarity of their node sets, using a threshold of 0.5. Each group with trips on at least three vehicle-days and two dates becomes a route, represented by its medoid trip. The fleet library holds 586 observed and 503 derived reverse routes. To limit the agent's decision space, the eligible routes of a driver are the smallest set of routes that covers 80% of its trip distance on qualified routes, together with their reverse routes. Routes shorter than 5 km in the network are excluded. A vehicle with at least one eligible route qualifies for the models: both models electrify it, and it becomes an EV agent in M_A. In the selected communities, 89 of the 101 vehicles qualify in Khayelitsha, all 44 in Bellville and 20 of the 26 in Wynberg (Table 7). In M_A, a route is one discrete unit of operation, where a driver cannot charge at a rank part-way along it.

4.2.3. Vehicle communities and pilot selection

Vehicle communities are identified from four views of observed operation: dwell time at network nodes, dwell time at formal ranks, use of network edges, and origin–destination pairs of anchor dwells. Features are weighted using term frequency–inverse document frequency (TF–IDF). The similarity between two vehicles is calculated as the mean cosine similarity across the views available for both vehicles.

Each vehicle is connected to its 15 nearest neighbours in a similarity graph. The Leiden algorithm partitions this graph across a range of resolution parameters, using 50 random seeds at each resolution. The seed-level partitions are combined into a consensus partition. A resolution of 0.75 is selected, producing eight communities while providing the best stability among the resolutions that retain at least 90% of corridor use within a single community.

Candidate charging sites are identified from each community's most frequently used non-home nodes. Nodes are ranked by the number of community vehicles with dwell periods of at least 20 minutes, and the first three are retained as candidate sites in both models. Each community is then screened using the energy core of M_R under three charging scenarios: overnight charging only; overnight charging plus unrestricted 60 kW charging at the three candidate hubs; and hub charging only, without overnight charging. The corresponding viable shares of vehicle-days are denoted by V_1, V_2 and V_3. The binding of hub charging, $B = V_2 - V_1$, is the gain in the viable share when hub charging is added to overnight charging. A large binding means that overnight charging alone leaves many vehicle-days non-viable, so the community needs hub chargers. For each candidate hub, the estimated connector requirement is based on the 95th percentile of the observed daily peak number of simultaneous charging sessions in the second scenario. This screening rule was established to avoid biasing the outcomes of the data-replay agent-based comparison.

A community qualifies for pilot selection when it contains at least 15 vehicles, has a formal main hub, has a binding of hub charging of at least 15 percentage points ($B \geq 15$), and has eligible routes that represent at least 30% of the median driver's trip distance. The last condition ensures that the agent-based model represents a meaningful share of community operation. The three qualifying communities with the highest V_2 values are selected, subject to having distinct main hubs. Consequently, the results apply most directly to communities that meet these electrification-readiness criteria, rather than to all minibus-taxi communities.

4.3. Fixed-mobility data-replay

The data-replay model (M_R) follows the trajectory-replay approach used in earlier minibus-taxi electrification studies [29, 30]. Each valid vehicle-day of the qualified vehicles on the term weekdays that is not a long-distance day is replayed using the energy-consumption and battery assumptions of the study EV [7, 68]. Trajectories, departure times and dwell periods remain fixed. A vehicle can charge when it is parked near an available charger but cannot divert from its observed movement to reach one.

A charging opportunity is defined as a stop lasting at least 10 minutes within 100 m of a candidate charging site. Vehicles from each date, called a fleet day, are replayed in chronological order, sharing the available connectors. A vehicle whose battery is below $0.9C$ joins the first-in, first-out queue and charges at the connector power (P) of the design and charger efficiency (η) until either its battery reaches $0.9C$ or the observed stop ends. Since the replay uses observed dwell periods, a vehicle can charge during time spent waiting in a rank loading queue. This assumes that charging connectors are available at the loading bays. The agent-based model does not permit this as a driver must leave the loading queue to travel to a charging station. This asymmetry is retained in the comparison and favours the replay model. Additionally, every vehicle can also charge overnight, subject to the available parked time and the upper battery limit. The energy added overnight is

$$E^{\text{night}} = \min\left(0.9\,C - E^{\text{end}},\ \eta P_o\,\Delta t_{\text{park}}\right), \qquad \Delta t_{\text{park}} = t^{\text{start}} + 24\,\text{h} - t^{\text{end}}, \tag{2}$$

where $P_o = 3.7$ kW is the overnight charging power, E^{end} is the battery energy at the end of driving, and t^{start} and t^{end} are the start and end of the driving window.

Following Pretorius et al. [29], a vehicle-day is viable when the vehicle can repeat its duty cycle in the steady state without its battery falling below the $0.2\,C$ reserve. Each date is therefore simulated repeatedly, starting at $0.9C$, until the daily starting battery levels converge to a fixed point or a sustainable cycle within a tolerance of $0.001C$, all active vehicles have been resolved, or 100 iterations have been completed [30]. A vehicle that reaches zero battery energy strands and leaves the active set. Each vehicle-day is classified as viable, below reserve, stranded or unconverged. A below-reserve or stranded day is non-viable, and a day that is not unconverged is conclusive. The stranded share is also reported for M_A, where stranding is treated as a failure condition, providing a common reliability measure for the comparison.

4.4. Decision-adaptive agent-based model

The agent-based model (M_A) is implemented using the discrete-event simulation framework established by Sello and Booysen [46]. Each qualified vehicle is represented by a driver agent, called an EV agent once it is electrified, who decides when to work, which route to operate, how long to wait for passengers, when to idle and, after electrification, when to charge. Unlike the replay model, M_A generates vehicle movement from these decisions.

Each simulated day represents a typical term weekday on the inferred operational network. The model specification follows the Overview, Design concepts and Details (ODD) protocol [70, 71], with the decision-making questions of its ODD+D extension [72], and the full description is provided in Appendix C. The following subsections focus on the mechanisms relevant to the comparison and on changes made relative to the earlier model.

4.4.1. Operation

When a driver is not servicing a route, the agent chooses between its eligible routes and idling using a multinomial logit model [73]. Route utility depends on the requests waiting at the current node, the number of competing drivers, and the driver's memory of route-level demand. This memory records the number of passengers served and their passenger-distance contribution during each 30-minute interval of the day. Idling becomes more attractive during intervals that the driver remembers as having low demand.

At each stop, the driver decides whether to continue waiting for more passengers, the hold-back of Section 3, according to

$$\pi = \left[1 + \exp\left(-\left(b + w_d\,d - w_o\,o\right)\right)\right]^{-1}, \tag{3}$$

where d is the remembered demand for the route, scaled by vehicle capacity, and o is the current onboard occupancy as a share of capacity. The parameters b, w_d and w_o are calibrated. The initial stop of a route

represents fill-and-go loading at a rank and has separate parameters, allowing drivers to wait according to their remembered demand at the terminal.

Relative to the earlier model by Sello and Booysen [46], the stop decision now uses route-specific demand memory, rank loading is represented explicitly, and includes a last-run rule that prevents a driver from starting a route that would extend beyond its shift. A passenger boards only if the vehicle is at the request's origin, has an available seat and serves the passenger's destination. Unserviced requests expire after two hours. Travel times are drawn from windowed log-normal distributions fitted to observed network hops, while each agent's shift is based on its observed first and last movements.

4.4.2. Charging

During the electrification experiments, the EV agents are electrified, while their ICE competitors remain internal combustion vehicles. Each EV driver follows a demand-aware charging policy. The vehicle always charges when its state of charge falls below $s_{\min} = 0.2$, never charges above $s_{\max} = 0.9$, and charges between these limits when

$$\frac{E(t)}{C} < s_{\min} + (s_{\max} - s_{\min}) \left(1 - \hat{h}_t\right), \tag{4}$$

where $E(t)$ is the battery energy and $\hat{h}_t \in [0, 1]$ is the normalised demand remembered by the driver for the current time interval. The policy makes charging more attractive when the driver expects charging to incur a smaller passenger-service opportunity cost.

Drivers can initiate charging only when they are not on a route. They travel to the nearest charging station and join its first-in, first-out queue. A driver waiting to load at a rank must leave the queue to charge and subsequently return to passenger service. Reserve rules reduce avoidable stranding: a driver may start a route only when the battery can support the route, the subsequent drive to the nearest station and the reserve of $0.2C$. The reserve is the minimum level $s_{\min}$ of the charging policy times the capacity, so a change of $s_{\min}$ also changes the reserve. A route is not eligible if a full battery cannot support its completion. While a vehicle is at home, it can charge at 3.7 kW. If an EV strands, it loses its onboard passengers and restarts from home with a full battery on the following day.

4.4.3. Run horizon

A cold simulation begins with empty driver memories and includes a warm-up period of 49–91 days. The warm-up ends when the seven-day means of passenger-service and route-choice measures remain close to their long-run means for three random seeds. The accepted warm-up is stored and reused for each experiment within a community. Each experiment begins from this saved warm start and runs for 14 settling days, allowing EV drivers to learn their charging behaviour. These are followed by 14 evaluation days, which provide the outcomes used for design selection and comparison.

4.5. Mixed-fleet population and passenger demand

The GPS dataset covers only a subset of the taxis operating from each hub — those fitted with Sebenza Wi-Fi routers. Thus, to represent all the drivers on these routes, M_A adds ICE competitors: synthetic copies of the EV agents that follow the same decision model. The resulting population is, on average, k times the number of EV agents. Each copy follows its associated agent's eligible routes, idle locations and shift pattern, but is assigned a nearby home location and a shift adjusted by a random draw with a standard deviation of 30 minutes. This construction assumes that the EV agents provide a useful representation of the wider population operating from the same hub. The EV agents account for 2.6%–6.6% of the simulated driver population (Table 7), reflecting the small share expected in an initial pilot. The ICE competitors retain ICE drivetrains and compete with the EVs for passenger requests.

Since GPS trajectories do not reveal passenger demand directly, the model generates a latent demand process using observed operating patterns and published passenger counts. The observed departures, routes and stops determine the structure, direction and timing of demand, while daily boardings at the main hub in the CITP [59] determine its overall scale. Each departure has a load factor of 1, except for counter-peak travel during the morning and afternoon peaks. For those departures, the load factor is set to the counter-peak-to-peak ratio of vehicle trips in CITP Table 3-18, which ranges from 0.09 to 0.39. The demand scale factor is

$$k = \frac{B}{Q\,L}, \tag{5}$$

where B is the published daily number of boardings at the main hub, Q is vehicle capacity and L is the estimated number of full agent loads from the hub per calibration day.

A share of 0.8 of passengers board at the route origin and alight downstream in proportion to the observed stop-event distribution. The remaining passengers travel between intermediate route nodes. Demand is generated only for origin–destination pairs that are served by at least one eligible route, follows the hourly operating profile of the EV agents and is sampled as a non-homogeneous Poisson process. This construction makes the simulated terminal boardings at the main hub equal the published boarding total by design.

The scale factor k is checked against an upper bound inferred from the provincial operating-licence register [65]. The Bellville estimate exceeds this bound, indicating uncertainty in the relationship between the qualified vehicles and the full hub population. This limitation is discussed in Section 6.4; it should not be interpreted as evidence that the synthetic population is an exact reconstruction of the entire taxi service.

4.6. Calibration and validation of the agent-based model

Calibration fits the behavioural parameters of M_A to the observed movement. Validation then checks the calibrated model against what the fit did not use: cold runs on new seeds, the two hold-out days and the quantities of the replay rules (Section 4.6.2). In the design experiment, a validated design has a different meaning: a search candidate that is evaluated again on the selection seeds (Section 4.7.2).

4.6.1. Calibration

The ICE baseline of each community is calibrated to reproduce observed operating patterns that determine an EV's daily energy requirement and opportunities for charging. Calibration targets include the 10th, 50th and 90th percentiles of daily distance, daily moving time and each agent's day-to-day coefficient of variation in daily distance. The hourly share of agents moving is also assessed, but is used as a realism check rather than a target that must be matched hour by hour. Additionally, the model must serve at least 95% of passenger requests, and median seat occupancy should approach 0.6. These are plausibility constraints and goals rather than observed calibration targets. The fit of each target is expressed as its absolute error relative to the target's estimated sampling variability, calculated from 500 bootstrap resamples of complete calibration dates.

Following Thiele et al. [47], calibration proceeds in two stages. Hand tuning first identifies a plausible region of parameter space and reveals structural gaps that cannot be resolved by parameter adjustment alone. In Khayelitsha, these gaps required a travel-time scaling adjustment and the introduction of the last-run rule. A predefined search strategy then optimises nine behavioural parameters using a constrained, two-objective TPE (Section 4.1.4). Selected configurations are validated with cold runs, which start from empty driver memories and run the full warm-up, on three new random seeds.

In Khayelitsha the search started from the calibrated values of Wynberg, and no validated search trial fitted the daily distance and moving time better than this start point, so the start point was accepted (Appendix D). Finally, the accepted configuration is assessed against the held-out observation days.

4.6.2. Fidelity of the simulated movement

The movement generated by M_A differs from the observed movement as agents operate on a simplified network, choose from a limited set of inferred routes and stop only at network nodes. Differences between the selected charging designs could therefore arise from the representation of movement, rather than from driver adaptation alone. To assess this effect, the data-replay rules are applied to the simulated movements from five validated ICE runs of M_A. Each simulated home-to-home duty is treated as one vehicle-day. Distance is calculated from network travel, charging opportunities are identified from simulated dwell periods at candidate sites, and overnight energy is calculated using the same rule as in Equation (2). The first eight evaluation days of each seed are used to match the number of days replayed in M_R, and outcomes are averaged across the five seeds.

The resulting charging design, $\mathbf{x}_S^\star$, is the design selected by the M_R rule when applied to simulated rather than observed movements. Comparing $\mathbf{x}_R^\star$ with $\mathbf{x}_S^\star$ isolates the effect of the movement representation under the replay selection rule. Comparing $\mathbf{x}_S^\star$ with $\mathbf{x}_A^\star$ captures driver adaptation together with the differences

between the models' service measures and selection rules. The difference in EV passenger change between $\mathbf{x}_R^\star$ and $\mathbf{x}_A^\star$ is decomposed as

$$\underbrace{\bar{\Delta}_{\mathrm{EV}}(\mathbf{x}_R^\star) - \bar{\Delta}_{\mathrm{EV}}(\mathbf{x}_A^\star)}_{\text{total}} = \underbrace{\bar{\Delta}_{\mathrm{EV}}(\mathbf{x}_R^\star) - \bar{\Delta}_{\mathrm{EV}}(\mathbf{x}_S^\star)}_{\text{representation}} + \underbrace{\bar{\Delta}_{\mathrm{EV}}(\mathbf{x}_S^\star) - \bar{\Delta}_{\mathrm{EV}}(\mathbf{x}_A^\star)}_{\text{behaviour}}, \tag{6}$$

where $\bar{\Delta}_{\mathrm{EV}}$ is the mean EV passenger change. The fidelity of the movement is reported with the quantities that the replay rules use: the daily distance, the overnight time, the charging opportunities at the sites and the ceiling of the not-stranded share. No pass threshold is set.

4.7. Charging design experiment

4.7.1. Candidate sites and design space

Each charging design is represented by

$$\mathbf{x} = (N_1, N_2, N_3, P) \in \mathcal{X}, \qquad N_j \in \{0, 1, \ldots, 2\hat{N}_j\}, \qquad P \in \{22, 50\}\ \mathrm{kW}, \tag{7}$$

where N_j is the number of connectors installed at candidate hub j, and P is the rated power of each connector. Each hub can have between zero and $2\hat{N}_j$ connectors, where $\hat{N}_j$ is the connector estimate derived during screening. Connector power is either 22 or 50 kW. A value of $N_j = 0$ means that no charging site is installed at hub j. Site selection is therefore part of the optimisation. The resulting design space $\mathcal{X}$ contains 2,645 candidates in Khayelitsha, 1,169 in Bellville and 349 in Wynberg. Table 13 gives $\hat{N}_j$ for each hub. To test whether the bound constrains the selection, three designs above it, up to $3\hat{N}_j$, are also evaluated in M_A in Khayelitsha, the community with the largest selected designs.

4.7.2. Selection rules

Each model selects its design with the measures of Section 4.1.4.

The replay model is deterministic, allowing every candidate design to be evaluated. Its Pareto front is therefore exact within the specified design space. Because replayed vehicles cannot divert to a charging station, even unlimited connectors cannot prevent every vehicle-day from stranding. The ceiling is therefore the highest not-stranded share of the enumerated designs, which is the service of the best design. A design with 99 connectors at each candidate site confirms that the design space reaches this ceiling. The replay candidates must reach at least 95% of the ceiling. Among those candidates, $\mathbf{x}_R^\star$ is selected by minimising peak site load and then installed power. The selected design is used as the common reference for the first research question.

The agent-based model is more computationally expensive, so it cannot feasibly evaluate every candidate design. Instead, a multi-objective TPE search proposes 40 trials using one seed per trial, starting from $\mathbf{x}_R^\star$. A design that the search proposes again reuses its first result, so the 40 trials evaluate 36 to 39 distinct designs (Table 20). The search minimises expired passenger requests, mean passenger waiting time and peak site load, while maximising the passengers carried by the EV agents. The resulting candidates and $\mathbf{x}_R^\star$ are then evaluated on five new seeds, the selection seeds, with their paired ICE counterfactuals; a design evaluated on these seeds is called a validated design. Unlike the replay front, the agent-based front is therefore an estimate. The final agent-based design, $\mathbf{x}_A^\star$, is chosen with the relative tolerance of the replay rule. With $S = 1 + \Delta_{\mathrm{EV}}$, a candidate must keep at least 95% of the EV passengers of the best validated design, $S \geq 0.95\, S_{\mathrm{best}}$, and its mean expired-request share must be within one percentage point of the minimum. Among the candidates, the selected design has the lowest peak site load and then the lowest installed power. No design is selected manually. The two models therefore use the same form of tolerance. The selection is also reported for tolerances of 90%, 97.5% and 99%, and for the knee of each front, the front design nearest to the origin when each objective is scaled to its range on the front. A validated EV run that strands invalidates the selection; this condition applies to the selection runs only.

4.7.3. Cross-evaluation, sensitivity and synthesis

The consequences of selecting infrastructure under fixed-mobility assumptions are measured by comparing the outcomes of $\mathbf{x}_R^\star$ and $\mathbf{x}_A^\star$ in M_A on ten seeds that the search and the selection did not use. On the selection seeds, part of the lead of $\mathbf{x}_A^\star$ is seed noise, so a comparison on these seeds would favour $\mathbf{x}_A^\star$. For each outcome,

the paired difference is calculated between the two designs on each seed and then averaged. The standard error of the mean paired difference quantifies the uncertainty across the ten seeds. Since both designs use the same seed and warm start, the paired comparison reduces variation arising from different random simulation conditions. Differences within two standard errors of zero are treated as indistinguishable at the resolution of this experiment.

Changes in expired requests are also examined. The share of the passengers that the EV agents lose and the ICE competitors take is measured directly. For each EV run and the ICE run of the same seed, the capture share is the change of the passengers of all other drivers (gained) divided by the passengers that the EV agents lose (lost). It can exceed 1 when the other drivers gain more passengers than the EV agents lose, through seed variation or because fewer requests expire.

Robustness is assessed by evaluating $\mathbf{x}_R^\star$ and $\mathbf{x}_A^\star$, without reselecting either design, under six alternative conditions: energy consumption 15% below and above the baseline; charging-policy minimums of 0.1 and 0.35; and fleet scale factors k equal to 0.5 and 1.5 times their baseline values. Each case is evaluated on five fresh seeds, with a corresponding ICE counterfactual where the changed condition affects ICE operation.

For each community, the analysis records three results: the difference in installed power between $\mathbf{x}_A^\star$ and $\mathbf{x}_R^\star$, the cost of $\mathbf{x}_R^\star$, and the EV passenger change achieved by $\mathbf{x}_A^\star$ relative to the ICE counterfactual. The cost of $\mathbf{x}_R^\star$ is the number of percentage points by which the EV passenger change of $\mathbf{x}_R^\star$ falls short of that of $\mathbf{x}_A^\star$ when both are evaluated in M_A. The tables report the paired difference, $\mathbf{x}_R^\star$ minus $\mathbf{x}_A^\star$, which is the negative of this cost. Because each model selects its design in its own objective space, the direction of the cross-evaluation in M_A is partly built into the method. The analysis therefore reports the size of each difference and its form: whether the agent-based design adds capacity, moves connectors between sites, or both. A result is robust when its size and form persist across the tested cases. These experiments assess the robustness of the selected designs' outcomes. They do not test whether the designs themselves would remain optimal under the alternative conditions.

To relate the passenger result to an operator's decision, each design is also priced (Table 21). The annual cost is the installed connector cost, annualised over ten years, plus the electricity demand charges of each site. A 50 kW DC connector costs the hardware and the installation cost of a US estimate for its number of chargers at the site [74], corrected from 2019 to December 2025 with the US producer and consumer price indices and converted at the December 2025 exchange rate. The demand charges are those of the City of Cape Town low-voltage time-of-use tariff of 2025/26 [75]: a charge on the measured peak of each site and a network capacity charge on its installed power, at a power factor of 0.96. The revenue is the fare of the additional passengers that the EV agents carry, on 250 term weekdays a year. Because the fare of the studied routes is not published, the result is the break-even fare at which the additional fare revenue pays the additional cost of $\mathbf{x}_A^\star$ against $\mathbf{x}_R^\star$. The discount rate (0, 4, 8 and 12% real) and the operating days (250 and 300) are varied.

5. Results

5.1. Selected communities and model validation

5.1.1. Communities

The community-detection procedure identifies eight vehicle communities (Table 6). Hub charging improves viability in every community, but the attainable viable share remains limited because the fleet's median daily distance of 190 km exceeds the 130 km of the EV's battery window. The predefined screening rule selects Bellville (V_2 = 73.6%), Wynberg (V_2 = 50.3%) and Khayelitsha (V_2 = 36.9%). Three communities fail at least one eligibility criterion, while two eligible communities rank below the selected three.

The selected communities differ in how well their operation is represented by the inferred route sets and network. Their median route coverage ranges from 36.9% to 62.9% of trip distance, while the share of driving outside the operational network ranges from 18.6% to 41.0% (Table 7). These differences are important when interpreting the agent-based results, because its drivers are restricted to the inferred network and their eligible routes.

5.1.2. Calibration

The calibrated ICE baselines reproduce the central parts of the observed duty-cycle distributions that determine EV energy use and charging opportunities (Table 8). The differences between observed and simulated median daily distance are 2.3, 12.9 and 1.4 km in Khayelitsha, Bellville and Wynberg, respectively. The

Table 6: Electrification screening of each community and the reason for each exclusion. V_1, V_2 and V_3 are the shares of the screened vehicle-days that are viable with overnight charging only, with overnight and hub charging at the first three hubs, and with hub charging only. Binding: the binding of hub charging, $B = V_2 - V_1$ in percentage points, the gain in the viable share when hub charging is added to overnight charging; a community is eligible only when $B \geq 15$. A main hub is formal when it is a rank, a station or a terminus.

Community	Vehicles	Main hub	V_1 (%)	V_2 (%)	V_3 (%)	Binding (pp)	Selected	Reason
C1 Khayelitsha	101	Rank	14.4	36.9	5.9	22.5	Yes	–
C2	65	Terminus	15.1	34.4	2.6	19.3	No	A higher-ranked candidate was selected
C3	51	Rank	15.5	34.4	8.6	18.9	No	A higher-ranked candidate was selected
C4 Bellville	44	Rank	23.9	73.6	2.5	49.7	Yes	–
C5 Wynberg	26	Terminus	21.9	50.3	4.7	28.4	Yes	–
C6	17	Terminus	31.5	51.4	3.6	19.8	No	Routes cover too little of the driving
C7	17	Terminus	55.0	70.6	15.6	15.6	No	Routes cover too little of the driving
C8	14	Rank	45.5	71.6	9.1	26.1	No	Fewer vehicles than the minimum

Table 7: The selected communities, their data and their configured models. Qualified: the vehicles with an eligible route of 5 km or more, which both models electrify. Drivers: the EV agents and their ICE competitors in M_A, with the scale factor k. Routes are the distinct configured routes of the drivers. Valid: the valid share of the community's vehicle-days. Off net.: the share of the driving outside the network. Route cov.: the median share of a vehicle's trip distance on its eligible routes, over the 93, 44 and 25 vehicles with an eligible route before routes shorter than 5 km were excluded.

Community	Vehicles	Qualified vehicles	Drivers	k	Routes	Requests a day	Valid (%)	Off net. (%)	Route cov. (%)
Khayelitsha	101	89	1,358	15.21	395	78,561	67.8	32.5	36.9
Bellville	44	44	1,673	38.03	259	130,881	68.6	18.6	62.9
Wynberg	26	20	686	34.30	114	43,172	61.6	41.0	41.0

corresponding differences in median daily moving time are 0.3, 0.2 and 0.5 hours. In all three communities, the model serves at least 95.2% of passenger requests.

The main remaining discrepancies occur in the distribution tails. In Wynberg, the 10th percentile of simulated daily distance is 91 km, compared with 113 km in the observations. The simulated agents also vary less from day to day than the observed vehicles: the 90th percentile of the distance coefficient of variation is 0.26–0.35 in the simulation, compared with 0.37–0.75 in the observations. The simulated hourly activity profiles also have lower peaks and relatively more activity around midday. Despite these differences, the accepted configurations meet the predefined hold-out criteria. The hold-out gaps are 0.83, 0.58 and 0.65 in Khayelitsha, Bellville and Wynberg, respectively, all below their corresponding acceptance limits of 1.31, 1.69 and 1.75. This supports the use of the calibrated models for the planned comparisons, while the residual differences in activity patterns remain relevant to their interpretation.

5.1.3. Agreement between observed and simulated movement

Table 9 compares observed vehicle movements with the movements generated by the calibrated ICE baselines of M_A, using the quantities required by the replay rules. Daily distance is broadly consistent between observed and simulated movements in all three communities, suggesting reasonable agreement in total daily energy requirements under the assumed consumption rate.

However, two differences remain. First, simulated overnight periods are 1.5–2.5 hours longer at the median than observed periods. This difference provides the simulated EVs with an additional 5–8 kWh of potential home-charging energy per day, equivalent to 6%–15% of the median daily energy requirement. Second, the frequency of charging opportunities at candidate sites differs across communities. In Khayelitsha, the simulated agents stop at candidate sites about half as often as the observed vehicles, reducing the ceiling of the not-stranded share from 45.1% to 31.0%. In Bellville, simulated agents stop at candidate sites more often than observed vehicles. In Wynberg, the observed and simulated movements provide similar numbers of site opportunities. Figure 3 presents the full distributions.

Table 8: Duty-cycle fit of the baseline agent-based model of each community. Obs.: the calibration target from the vehicle-days of the qualified vehicles on the calibration days. M_A: the run of the accepted configuration from its warm start. M_R replays the observed movement, so it reproduces the observed values by construction and is not shown. The served share (constraint: at least 0.95) and the seat occupancy (goal: up to 0.6) are plausibility checks with no observed value.

	Khayelitsha		Bellville		Wynberg	
Measure	Obs.	M_A	Obs.	M_A	Obs.	M_A
Daily distance, P10 (km)	126.4	120.2	110.3	114.0	113.1	91.0
Daily distance, P50 (km)	212.6	210.3	150.5	163.4	200.5	201.9
Daily distance, P90 (km)	323.4	317.9	226.7	231.6	298.7	318.4
Moving time, P10 (h)	5.20	4.82	3.74	3.65	4.34	2.99
Moving time, P50 (h)	8.27	8.58	5.20	5.05	6.04	6.58
Moving time, P90 (h)	11.47	12.21	7.52	6.83	9.58	9.55
Daily distance CV, P10	0.04	0.13	0.09	0.13	0.08	0.14
Daily distance CV, P50	0.15	0.20	0.14	0.17	0.14	0.20
Daily distance CV, P90	0.75	0.34	0.37	0.26	0.73	0.35
Served requests (share)	–	0.958	–	0.952	–	0.967
Seat occupancy on route runs	–	0.39	–	0.57	–	0.44

Table 9: Observed against simulated movement, measured with the quantities of the replay rules. Obs.: the replayed vehicle-days of the qualified vehicles on the eight term weekdays. Sim.: the home-to-home duties of their agents in the calibrated ICE runs of M_A (first eight evaluation days of five seeds). Distances and times are P50 [P10–P90]. A site opportunity is a stop of 10 min or more within 100 m of a candidate site. The battery window is $0.7\,C$. The ceiling is the highest not-stranded share that the replay rules reach over the design space; a design with 99 connectors at each site reaches no higher.

	Khayelitsha		Bellville		Wynberg	
Measure	Obs.	Sim.	Obs.	Sim.	Obs.	Sim.
Vehicle-days	463	3,424	234	1,697	99	773
Daily distance (km)	208 [127–309]	205 [116–321]	145 [109–220]	162 [105–232]	185 [119–288]	198 [98–320]
Overnight time (h)	6.9 [2.6–10.0]	8.4 [4.1–13.0]	5.7 [2.8–9.6]	8.2 [4.4–10.6]	5.7 [2.7–9.0]	7.8 [2.3–11.2]
Site opportunities per vehicle-day	2.44	1.22	4.08	6.75	1.99	1.70
Days without a site opportunity (%)	28.5	34.1	8.1	0.5	33.3	32.1
Days that need more than the window (%)	89.2	86.7	65.4	75.8	81.8	77.4
Ceiling of the not-stranded share (%)	45.1	31.0	80.8	84.6	53.1	47.5

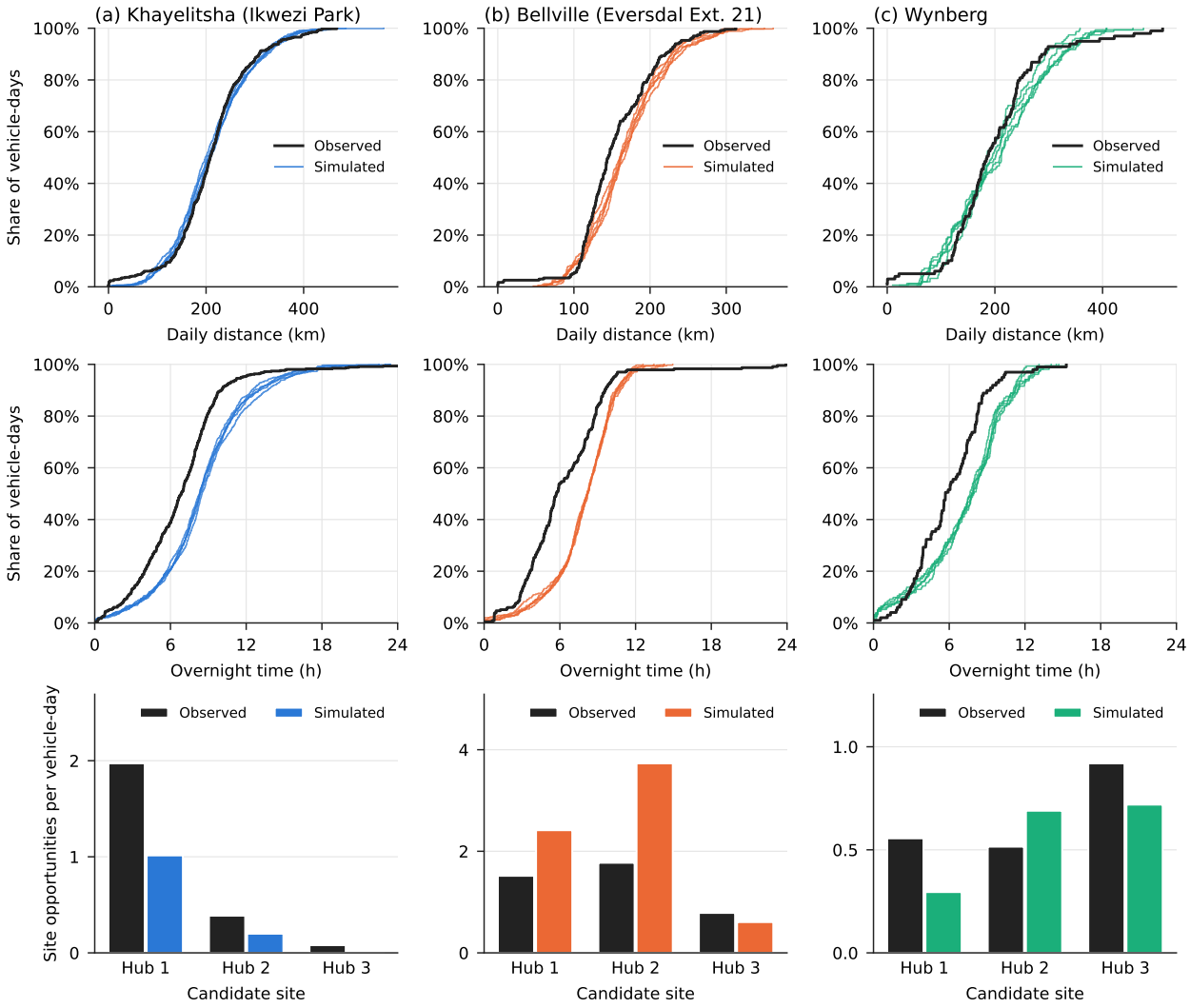


Figure 3: Observed against simulated movement in each community: (a) Khayelitsha, (b) Bellville, (c) Wynberg. Top and middle: the empirical cumulative distribution of the daily distance and of the overnight time. Observed: the replayed vehicle-days of the qualified vehicles on the eight term weekdays. Simulated: the home-to-home duties of their agents in the calibrated ICE runs of M_A, one line for each of the five seeds. Bottom: the site opportunities per vehicle-day at each candidate site, as in Table 9.

5.2. Electrification outcomes under a common design (RQ1)

Under the reference design $\mathbf{x}_R^\star$, the two models predict substantially different electrification outcomes (Table 10). In the data-replay model, 57.0%, 22.6% and 49.0% of vehicle-days strand in Khayelitsha, Bellville, and Wynberg, respectively. These failures largely reflect the mismatch between daily energy requirements and the charging opportunities available along the fixed trajectories. Median daily energy needs are 79, 55 and 70 kWh, whereas overnight charging at 3.7 kW supplies approximately 23, 19 and 19 kWh. Consequently, 94%–97% of vehicle-days require site charging to sustain repeated operation. Yet 29%, 8% and 33% of days, respectively, contain no qualifying stop near a candidate charging site. Even with unlimited connectors, the replay model therefore predicts that 55%, 19% and 47% of vehicle-days would still strand in the three communities (Table 9).

In the agent-based model, no EV run of the base case strands because drivers can travel to charging stations. This improved ability to obtain energy comes with a passenger-service cost. Under the same reference design, site energy use is 2.8, 1.7 and 4.0 times the replay values in Khayelitsha, Bellville and Wynberg, respectively. The model also records 2.4, 1.5 and 3.8 times as many site charging sessions. The EV agents nevertheless carry 18.1%, 6.9% and 26.4% fewer passengers than in their corresponding ICE counterfactuals.

For this common design, the models predict the same peak site load because all installed connectors are used at the peak. The replay model reports lower site energy partly because stranded vehicles stop charging and leave the active fleet. It reports much less home energy for three reasons. It replays only the valid vehicle-days, about 58, 29 and 12 a day against the 89, 44 and 20 EV agents of M_A; a stranded vehicle leaves the active fleet and no longer charges at home; and the simulated overnight periods are 1.5 to 2.5 h longer (Section 4.6.2). Figure 4 illustrates the differences in charging patterns, battery levels and the causes of replay failure.

Table 10: Outcomes of the selected M_R design of each community in both models. M_R: the replayed term weekdays; M_A: the mean over ten fresh seeds of 14 evaluation days each. Non-viable: the steady-state share of the M_R vehicle-days below the $0.2\,C$ reserve or stranded; M_A has no steady state. Low-battery: the share of the M_A EV agent-days on which a route run ends or a charging session starts below $0.2\,C$, where drivers look for a charger; it is not a failure. A dash shows a measure that the model cannot represent.

	Khayelitsha		Bellville		Wynberg	
Measure	M_R	M_A	M_R	M_A	M_R	M_A
Stranded EV-days (share)	0.570	0.000	0.226	0.000	0.490	0.000
Non-viable EV-days (share)	0.646	–	0.274	–	0.602	–
Low-battery EV agent-days (share)	–	0.006	–	0.004	–	0.016
Peak site load (kW)	250	250	200	200	100	100
Site energy (kWh a day)	1,368	3,791	1,020	1,778	299	1,198
Home energy (kWh a day)	356	2,102	323	1,081	129	435
Site sessions a day	99.1	239.5	83.1	120.5	16.4	62.4
Expired requests (share)	–	0.041	–	0.049	–	0.034
Mean passenger wait (min)	–	2.1	–	4.5	–	16.9
EV passenger change against ICE	–	−0.181	–	−0.069	–	−0.264
EV passenger loss ℓ	–	0.186	–	0.062	–	0.269

5.3. Selected charging infrastructure (RQ2)

The models select different charging designs in all three communities (Table 11 and Figure 5). The difference has two forms. In Khayelitsha, the agent-based model selects more capacity: 650 kW of installed power (5, 6 and 2 connectors at 50 kW; peak site load 600 kW on the ten fresh seeds), against 250 kW in the replay model. In Bellville and Wynberg, it selects the same installed power as the replay model (200 kW and 100 kW) at different hubs. In Bellville it puts all four connectors at the second hub, where the replay model puts one of them at the first hub. In Wynberg it puts both connectors at the first hub, where the replay model puts one at each of the first two hubs.

The agent-based search shows strongly falling returns to capacity in Khayelitsha. On the selection seeds, the EV passenger change rises from −64% at 22 kW to −18% at the replay design (250 kW) and −6% at the selected design (580 kW peak on these seeds). The designs with 660 to 860 kW of peak load gain only a further 0.6 to 3.4 percentage points. Under the replay model the same additional capacity has little effect:

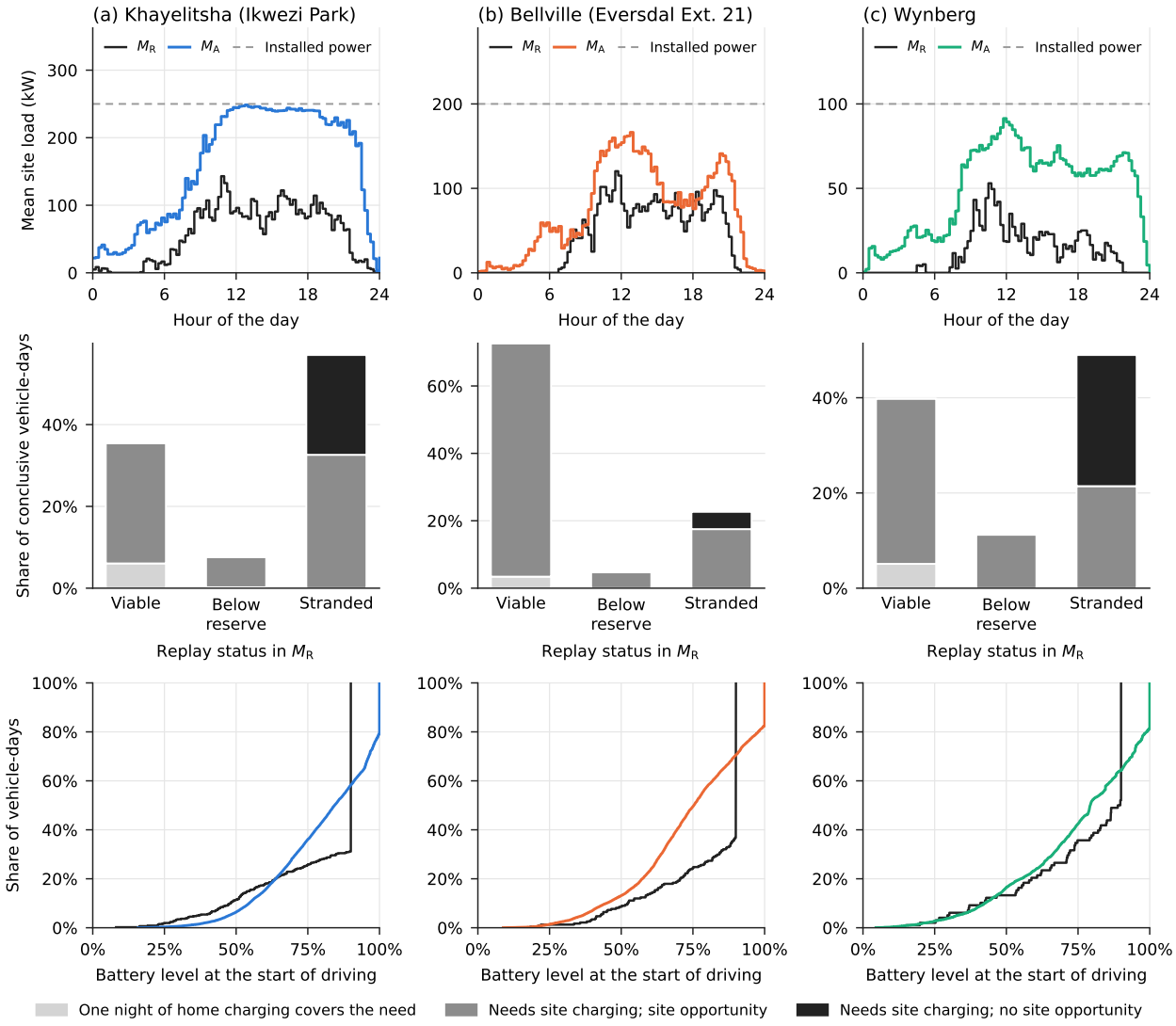


Figure 4: Charging under the reference design $\mathbf{x}_R^\star$ in both models: (a) Khayelitsha, (b) Bellville, (c) Wynberg. Top: the mean site load over the day in 15-minute intervals (M_R: the replayed term weekdays; M_A: ten fresh seeds of 14 evaluation days each) and the installed power (dashed). Middle: the replay status of the conclusive vehicle-days in M_R, divided by the condition of the day. A day needs site charging when one night of home charging at 3.7 kW does not cover its energy need. Bottom: the empirical cumulative distribution of the battery level at the start of driving (M_R: the start level of the replayed vehicle-day; M_A: the first logged level after the agent leaves home).

the selected agent-based design strands 55.1% of the vehicle-days against 57.0% for the replay design, because a replayed vehicle cannot divert to use the new connectors.

The selected capacity depends on the tolerance of the selection rule (Table 12). Both models accept a design that keeps at least 95% of the service of the best design. When this tolerance changes from 90% to 99%, the Khayelitsha agent-based design changes from 374 to 1,000 kW of installed power, and the replay design from 200 to 350 kW. At every tolerance the agent-based design has more installed power than the replay design in Khayelitsha, the same or less in Bellville, and the same in Wynberg. The knee of each front gives the cheapest efficient compromise: 150, 100 and 50 kW in the replay model and 286, 88 and 66 kW in the agent-based model.

The upper bound of the design space does not constrain the result. The selection rule is relative to the best validated design, the design with the highest EV passenger change on the selection seeds, so a bound that capped this design could also move the selection. In Khayelitsha, the best validated design is 19-8-6 (1,650 kW), which reaches the bound of $2\hat{N}_j$ at hubs 2 and 3 (8 and 6 connectors; Table 13). Three designs above the bound, up to $3\hat{N}_j$, do not improve the EV passenger change (differences of -0.1 to -1.0 percentage points, within two standard errors; Table E.28), so a larger design space would not change the best design or the selection; two of them give identical runs, because the drivers do not use the extra connectors.

Each design evaluated in both models (Table 14) shows that the two models disagree on which design is better in Bellville and Wynberg. There, the agent-based design strands more vehicle-days in the replay model (32.9% against 22.6% in Bellville, and 67.3% against 49.0% in Wynberg), while the replay design carries fewer EV passengers in the agent-based model. In Khayelitsha, the agent-based design is the better design in both models; the models disagree on whether its extra 400 kW is worth installing.

Table 11: Design that each model selects in each community: the connectors N_1, N_2 and N_3 at the first three hubs, the power P of a connector, the installed power and the peak site load. A design ID $n\,N_1$-N_2-N_3_$p\,P$ gives the same values.

Community	Model	Design	N_1	N_2	N_3	P (kW)	Installed (kW)	Peak (kW)
Khayelitsha	M_R	n03-01-01_p50	3	1	1	50	250	250
	M_A	n05-06-02_p50	5	6	2	50	650	600
Bellville	M_R	n01-03-00_p50	1	3	0	50	200	200
	M_A	n00-04-00_p50	0	4	0	50	200	200
Wynberg	M_R	n01-01-00_p50	1	1	0	50	100	100
	M_A	n02-00-00_p50	2	0	0	50	100	100

Table 12: Selected design of each model for each selection rule. Relative: among the designs that keep at least the fraction τ of the service of the best design, the design with the lowest peak site load, then the lowest installed power ($\tau = 0.95$ is the main rule of both models). Knee: the front design nearest to the origin when each objective is scaled to its range on the front. Designs are N_1-N_2-N_3 connectors of P kW. The M_A values are from the five selection seeds; the peak site load of M_R equals its installed power.

		M_R			M_A			
Community	Rule	Design	P	Installed (kW)	Design	P	Installed (kW)	Peak (kW)
Khayelitsha	$\tau = 0.90$	3-1-0	50	200	11-0-6	22	374	374
	$\tau = 0.95$	3-1-1	50	250	5-6-2	50	650	580
	$\tau = 0.975$	4-1-1	50	300	5-8-4	50	850	670
	$\tau = 0.99$	4-2-1	50	350	8-7-5	50	1,000	800
	Knee	2-1-0	50	150	6-3-4	22	286	286
Bellville	$\tau = 0.90$	1-2-0	50	150	1-2-0	50	150	150
	$\tau = 0.95$	1-3-0	50	200	0-4-0	50	200	200
	$\tau = 0.975$	1-3-1	50	250	0-4-0	50	200	200
	$\tau = 0.99$	2-3-1	50	300	0-4-0	50	200	200
	Knee	0-2-0	50	100	0-3-1	22	88	88
Wynberg	$\tau = 0.90$	1-1-0	50	100	2-0-0	50	100	100
	$\tau = 0.95$	1-1-0	50	100	2-0-0	50	100	100
	$\tau = 0.975$	1-1-1	50	150	2-0-1	50	150	150
	$\tau = 0.99$	1-1-1	50	150	2-0-1	50	150	150
	Knee	1-0-0	50	50	2-0-1	22	66	66

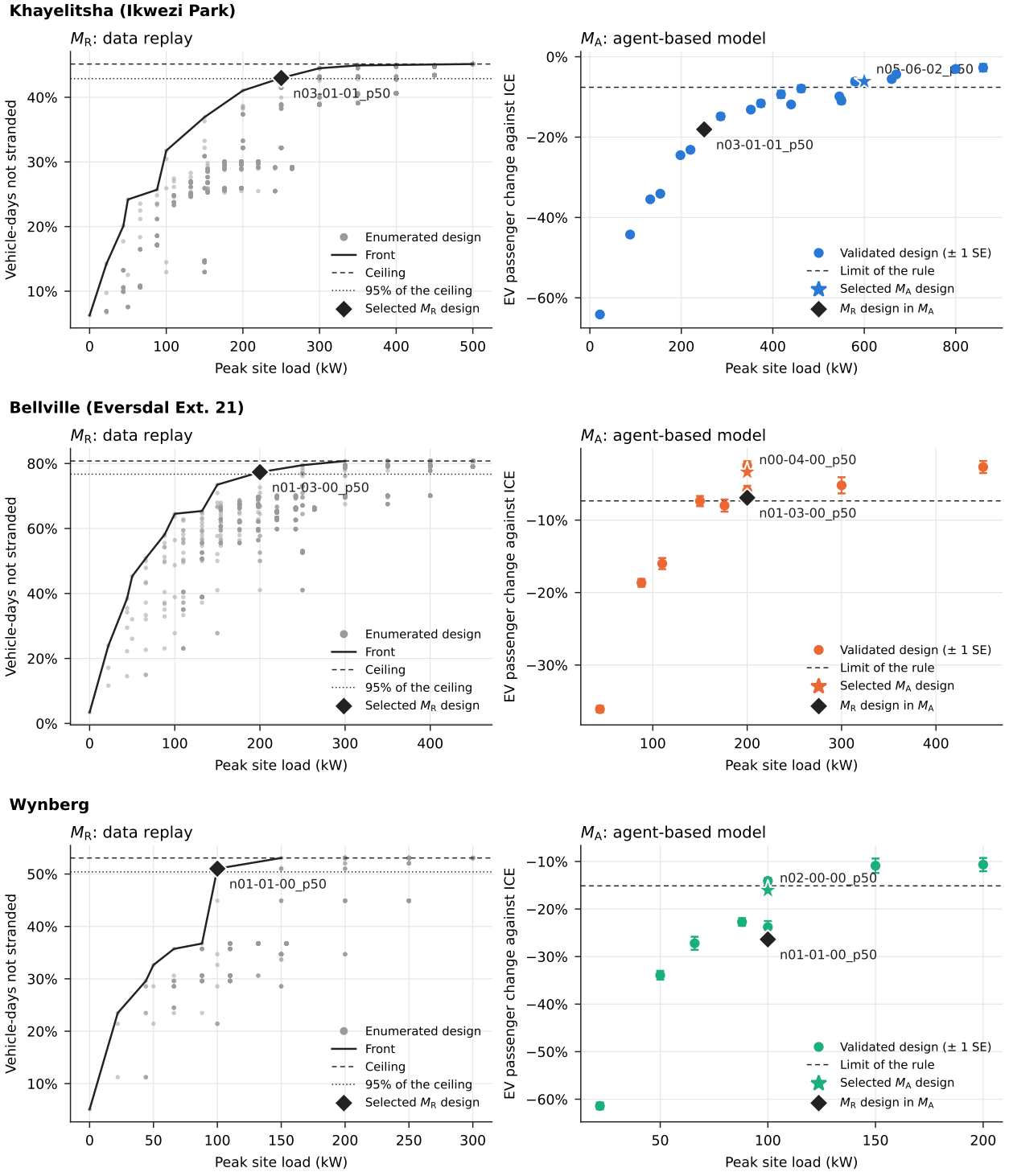


Figure 5: Design space of each community in both models, each in its model-native objectives: (a) Khayelitsha, (b) Bellville, (c) Wynberg. Left, M_R: every enumerated design (grey), the Pareto front (line), the ceiling of the not-stranded share (dashed) and 95% of it (dotted), and the selected M_R design (diamond). Right, M_A: the validated designs (mean ± one standard error over the five selection seeds), the 95% limit of the selection rule (dashed), the selected M_A design (star) and the M_R design evaluated in M_A (diamond).

Table 13: Connector estimate $\hat{N}_j$ and upper bound $2\hat{N}_j$ of each candidate hub, with the connectors of the selected designs. $\tau = 0.99$: the M_A design of the 99% tolerance where it differs from $\mathbf{x}_A^\star$.

Community	Hub	Type	$\hat{N}_j$	$2\hat{N}_j$	$\mathbf{x}_R^\star$	$\mathbf{x}_A^\star$	$\tau = 0.99$
Khayelitsha	1	Rank	10	20	3	5	8
	2	Terminus	4	8	1	6	7
	3	Stop	3	6	1	2	5
Bellville	1	Rank	4	8	1	0	–
	2	Stop	6	12	3	4	–
	3	Terminus	2	4	0	0	–
Wynberg	1	Terminus	2	4	1	2	2
	2	Rank	2	4	1	0	0
	3	Station	3	6	0	0	1

Table 14: Each selected design evaluated in both models. M_R: the stranded share of the vehicle-days and the peak site load on the replayed term weekdays. M_A: the EV passenger change against the ICE counterfactual (mean and standard error over ten fresh seeds) and the peak site load. $\tau = 0.99$: the M_A design of the 99% tolerance where it differs from $\mathbf{x}_A^\star$.

				M_R		M_A	
Community	Design	Connectors	Installed (kW)	Stranded (share)	Peak (kW)	EV passenger change (SE)	Peak (kW)
Khayelitsha	$\mathbf{x}_R^\star$	3-1-1	250	0.570	250	−0.181 (0.004)	250
	$\mathbf{x}_A^\star$	5-6-2	650	0.551	400	−0.061 (0.004)	600
	$\tau = 0.99$	8-7-5	1,000	0.549	500	−0.028 (0.005)	790
Bellville	$\mathbf{x}_R^\star$	1-3-0	200	0.226	200	−0.069 (0.005)	200
	$\mathbf{x}_A^\star$	0-4-0	200	0.329	200	−0.034 (0.009)	200
Wynberg	$\mathbf{x}_R^\star$	1-1-0	100	0.490	100	−0.264 (0.008)	100
	$\mathbf{x}_A^\star$	2-0-0	100	0.673	100	−0.161 (0.006)	100
	$\tau = 0.99$	2-0-1	150	0.653	150	−0.141 (0.013)	150

5.3.1. Representation and behaviour effects

The simplified movement representation does not explain the design difference (Table 15). When the replay selection rule is applied to simulated ICE movements, the resulting design $\mathbf{x}_S^\star$ differs from $\mathbf{x}_R^\star$ by at most 1.5 connectors moved between sites. On the fresh seeds, the representation part of the paired difference is +6.7 percentage points in Khayelitsha (SE 0.4), −1.6 in Bellville (0.7) and −0.1 in Wynberg (1.0), and the behaviour part is −18.6 (0.5), −2.0 (1.0) and −10.2 (1.0); a negative part adds to the cost of $\mathbf{x}_R^\star$. In Khayelitsha the representation works against the total: $\mathbf{x}_S^\star$ has one connector fewer than $\mathbf{x}_R^\star$ and loses more passengers. The difference between the selected designs therefore comes mainly from the operational responses of the drivers and from the selection criteria of M_A, not from the movement representation.

The decomposition combines behavioural effects with the differences in the models' service measures; it is not a causal estimate of behaviour alone.

Table 15: Representation and behaviour effects. $\mathbf{x}_S^\star$ is the design that the M_R rule selects on the simulated ICE movement of M_A. The step $\mathbf{x}_R^\star \to \mathbf{x}_S^\star$ isolates the representation effect; $\mathbf{x}_S^\star \to \mathbf{x}_A^\star$ contains the behaviour effect and the change of service measure and selection rule. Designs are N_1-N_2-N_3 connectors of 50 kW, with the installed power in kW. Change: the change of installed power in kW; moved: half the summed absolute change of the connectors per site. Difference: the EV passenger change in M_A of the first design minus that of the second, with its standard error over ten fresh paired seeds; the two parts sum to the paired difference in Table 16, the negative of the cost of $\mathbf{x}_R^\star$.

				Representation ($\mathbf{x}_R^\star \to \mathbf{x}_S^\star$)		Behaviour ($\mathbf{x}_S^\star \to \mathbf{x}_A^\star$)	
Community	$\mathbf{x}_R^\star$	$\mathbf{x}_S^\star$	$\mathbf{x}_A^\star$	Change (moved)	Difference (SE)	Change (moved)	Difference (SE)
Khayelitsha	3-1-1 (250)	3-1-0 (200)	5-6-2 (650)	−50 (0.5)	+0.067 (0.004)	+450 (4.5)	−0.186 (0.005)
Bellville	1-3-0 (200)	2-2-1 (250)	0-4-0 (200)	+50 (1.5)	−0.016 (0.007)	−50 (2.5)	−0.020 (0.010)
Wynberg	1-1-0 (100)	0-1-1 (100)	2-0-0 (100)	0 (1.0)	−0.001 (0.010)	0 (2.0)	−0.102 (0.010)

5.4. Cost of the replay design and the EV passenger loss (RQ3)

The replay-selected design $\mathbf{x}_R^\star$ is evaluated in M_A on ten seeds that the search and the selection did not use (Table 16). Its EV passenger change is 11.9, 3.5 and 10.3 percentage points lower than that of $\mathbf{x}_A^\star$ in Khayelitsha, Bellville and Wynberg (standard errors 0.4, 1.0 and 0.7). On the selection seeds the differences were 12.3, 3.5 and 9.7 points, within one point of the fresh-seed values in either direction, so selecting on these seeds did not inflate the lead of $\mathbf{x}_A^\star$. In Khayelitsha the difference comes with 400 kW of additional capacity. In Bellville and Wynberg the two designs have the same installed power and peak site load, so the whole difference of 3.5 and 10.3 points comes from the location of the connectors.

The EV passenger loss is not accompanied by an increase in the expired requests: the expired-request share changes by at most 0.04 percentage points, and the mean passenger wait by less than 10 s. The paired runs measure where the passengers go. In Khayelitsha the ICE competitors take 97% (SE 4%) of the passengers that the EV agents lose under $\mathbf{x}_R^\star$ and 85% (SE 11%) under $\mathbf{x}_A^\star$; in Wynberg they take 94% (SE 10%) and 98% (SE 17%) (Table 17). In Bellville the EV agents lose too few passengers (1,500 to 3,100 in 14 days) for the share to be determined against the variation of the other 1,629 drivers. Where the share is determined, the ICE competitors absorb almost all of the passengers that the EVs do not carry while they charge.

Even under $\mathbf{x}_A^\star$, EV passenger throughput remains below the ICE counterfactual. The time-adjusted EV passenger loss ℓ is 5.6% in Khayelitsha, 2.9% in Bellville and 16.4% in Wynberg. These remaining losses are the operational cost of electrification under the selected charging infrastructure and the modelled mixed fleet. No EV run of the base case strands. Figure 6 shows how the passenger losses vary over the day and how they relate to charging.

The time that the EV agents spend on charging explains the size and the form of the difference (Figure 6, bottom). Under $\mathbf{x}_R^\star$, they queue for a connector for 177, 52 and 223 min of each agent-day with a shift in Khayelitsha, Bellville and Wynberg, and under $\mathbf{x}_A^\star$ for 35, 13 and 94 min. The paired fall of the queue (142, 39 and 130 min, SE 2.6, 1.1 and 4.9) ranks the communities in the same order as their passenger gains. In Bellville and Wynberg, $\mathbf{x}_A^\star$ puts all connectors at one hub, and the queue falls by 75% and 58% at the same installed power. In Khayelitsha, the connectors of $\mathbf{x}_R^\star$ would need 15.2 h a day at full power to deliver their site energy (Bellville 8.9 h, Wynberg 12.0 h), against 7.3 h under $\mathbf{x}_A^\star$. The drive to a charger changes little with the design: about 30 min a day in Khayelitsha and Bellville, and 72 to 76 min in Wynberg.

Table 16: The selected M_R design $\mathbf{x}_R^\star$ against the selected M_A design $\mathbf{x}_A^\star$, both evaluated in M_A: means over ten fresh seeds, and the paired difference ($\mathbf{x}_R^\star$ minus $\mathbf{x}_A^\star$) with its standard error. The ICE run is the counterfactual of both designs. No EV-day strands in any run.

	Khayelitsha			Bellville			Wynberg		
Measure	$\mathbf{x}_R^\star$	$\mathbf{x}_A^\star$	Diff. (SE)	$\mathbf{x}_R^\star$	$\mathbf{x}_A^\star$	Diff. (SE)	$\mathbf{x}_R^\star$	$\mathbf{x}_A^\star$	Diff. (SE)
EV passenger change	−0.181	−0.061	−0.119 (0.004)	−0.069	−0.034	−0.035 (0.010)	−0.264	−0.161	−0.103 (0.006)
EV passenger loss ℓ	0.186	0.056	0.130 (0.005)	0.062	0.029	0.033 (0.009)	0.269	0.164	0.105 (0.007)
Expired requests (share)	0.041	0.041	0.000 (0.000)	0.049	0.049	0.000 (0.000)	0.034	0.034	0.000 (0.000)
Expired requests in the ICE run (share)	0.040		–	0.048		–	0.034		–
Mean passenger wait (s)	125.8	122.7	3.0 (1.2)	267.6	270.0	−2.5 (3.2)	1,016.8	1,007.7	9.1 (6.4)
Peak site load (kW)	250	600	−350 (10.5)	200	200	0 (0.0)	100	100	0 (0.0)
Site energy (kWh a day)	3,791	4,760	−968 (7.2)	1,778	1,922	−144 (11.6)	1,198	1,394	−197 (10.5)
Site sessions a day	239.5	316.3	−76.9 (1.9)	120.5	133.5	−12.9 (1.3)	62.4	78.6	−16.2 (0.4)

5.5. Size, form and robustness of the difference, and computational cost (RQ4)

Across the six sensitivity cases, evaluated on five fresh seeds, the cost of $\mathbf{x}_R^\star$ in M_A is 7.8 to 15.9 percentage points in Khayelitsha and 5.9 to 15.1 points in Wynberg, each at least four standard errors from zero (Figure 7 and Table 18). In Bellville it is 2.6 to 3.9 points in five cases, but with 15% lower energy consumption the two designs do not differ (+0.1 points, SE 1.2): the Bellville location effect exists only when the energy need is high enough to make the charging site matter.

Higher energy consumption widens the gap between the designs, and lower consumption narrows it. A lower charging-policy minimum also lowers the reserve of the route test, which is $s_{\min}C$; with $s_{\min} = 0.1$ the replay design strands on at most 0.17% (Bellville) and 0.36% (Wynberg) of the EV agent-days of a run, and with 15% higher energy consumption the Wynberg agent-based design strands on at most 0.37%. The rule that a stranded run invalidates the result applies to the selection runs only, where no run strands.

Table 17: Capture share: the share of the passengers that the EV agents lose and the other drivers take, from the EV run and the ICE run of the same seed. Lost: the passengers that the EV agents lose; gained: the change of the passengers of all other drivers; both are means over the seeds for the 14 evaluation days of a run. Capture share: gained divided by lost, as the mean over the seeds of the share of each seed. Fresh: the ten fresh seeds; selection: the five selection seeds. In Bellville, the passengers lost are few against the variation of the passengers of the other 1,629 drivers, so the capture share is not determined. A share above 1 means that the other drivers gain more passengers than the EV agents lose; every share above 1 is within two standard errors of 1.

		Fresh seeds			Selection seeds
Community	Design	Lost	Gained	Capture share (SE)	Capture share (SE)
Khayelitsha	$\mathbf{x}_R^\star$	12,503	12,145	0.97 (0.04)	0.86 (0.04)
	$\mathbf{x}_A^\star$	4,249	3,533	0.85 (0.11)	0.86 (0.18)
Bellville	$\mathbf{x}_R^\star$	3,139	2,130	0.71 (0.40)	1.07 (0.38)
	$\mathbf{x}_A^\star$	1,539	772	1.62 (1.78)	0.26 (0.70)
Wynberg	$\mathbf{x}_R^\star$	4,686	4,425	0.94 (0.10)	1.03 (0.17)
	$\mathbf{x}_A^\star$	2,852	2,827	0.98 (0.17)	0.57 (0.22)

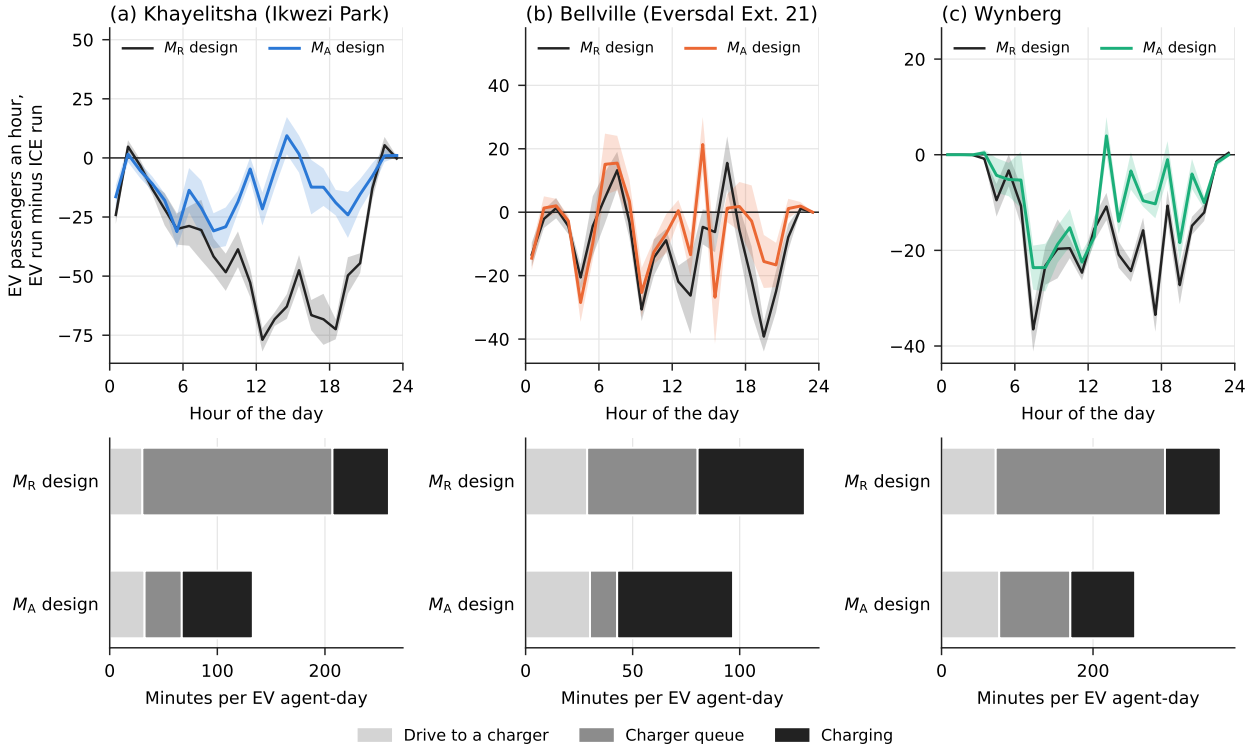


Figure 6: Passenger loss of the EVs over the day and the time that they spend on charging, under the replay design $\mathbf{x}_R^\star$ and the agent-based design $\mathbf{x}_A^\star$, both evaluated in M_A: (a) Khayelitsha, (b) Bellville, (c) Wynberg. Top: the passengers of the EV agents by the start hour of their route runs, minus those of the EV agents in their ICE counterfactual of the same seed (mean ± one standard error over ten fresh seeds). Bottom: the minutes for each EV agent-day on the drive to a charger, in the charger queue and charging.

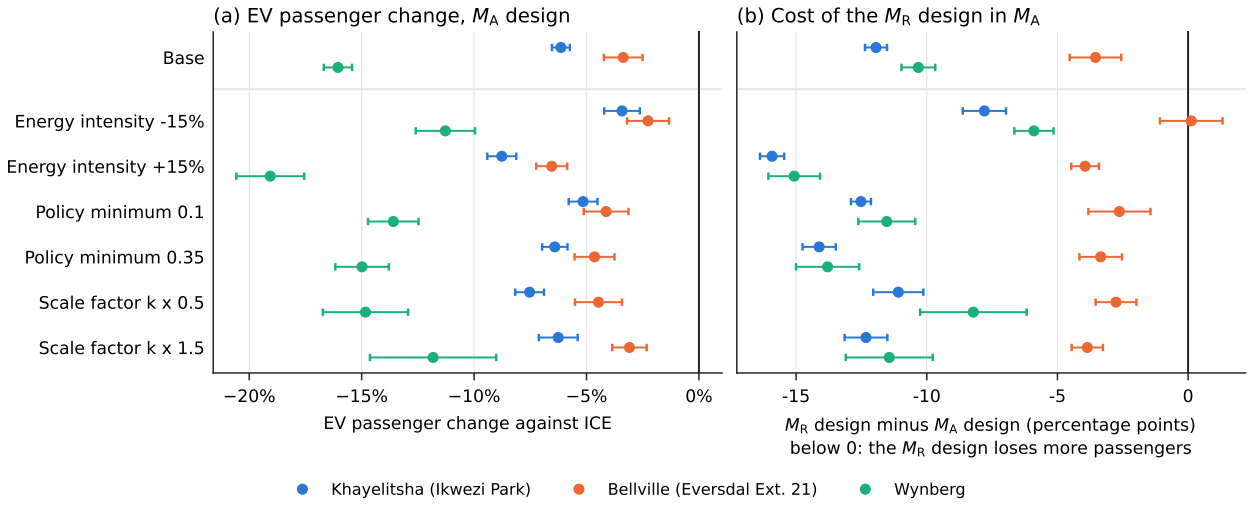


Figure 7: Sensitivity of the EV passenger change in M_A in each community (colour). (a) EV passenger change against internal-combustion operation of the selected M_A design. (b) EV passenger change of the selected M_R design minus that of the selected M_A design, both evaluated in M_A, in percentage points; below zero, the M_R design loses more passengers. Mean ± one standard error over five fresh seeds in each case and ten in the base case.

Table 18: EV passenger change against internal-combustion operation of the selected M_A and M_R designs, both evaluated in M_A, in each sensitivity case. Means over five fresh seeds; the base case uses the ten fresh seeds. The differences that the text gives are computed from unrounded values, so they can differ by 0.1 point from the rounded values of this table.

	Khayelitsha		Bellville		Wynberg	
Case	$\mathbf{x}_A^\star$	$\mathbf{x}_R^\star$	$\mathbf{x}_A^\star$	$\mathbf{x}_R^\star$	$\mathbf{x}_A^\star$	$\mathbf{x}_R^\star$
Base	−0.061	−0.181	−0.034	−0.069	−0.161	−0.264
Energy intensity −15%	−0.034	−0.112	−0.023	−0.021	−0.113	−0.172
Energy intensity +15%	−0.088	−0.247	−0.065	−0.105	−0.191	−0.341
Policy minimum 0.1	−0.052	−0.177	−0.041	−0.068	−0.136	−0.251
Policy minimum 0.35	−0.064	−0.205	−0.046	−0.080	−0.150	−0.288
Scale factor $k \times 0.5$	−0.075	−0.186	−0.045	−0.072	−0.148	−0.230
Scale factor $k \times 1.5$	−0.063	−0.186	−0.031	−0.069	−0.118	−0.233

The cross-community synthesis (Table 19) shows the form and the size of the difference. In installed power, only Khayelitsha differs (+400 kW); Bellville and Wynberg select the same power at different hubs. In EV passengers, the cost of $\mathbf{x}_R^\star$ is large in Khayelitsha and Wynberg (11.9 and 10.3 points) and small in Bellville (3.5 points), and the movement-representation component is small against it except in Khayelitsha, where it works in the opposite direction.

Table 19: Size and form of each difference between the models in each community: whether the two models select the same design, the installed power of the M_A design minus that of the M_R design, the form of the change (more capacity, other hubs, or both), and the EV passenger change of the M_R design minus that of the M_A design in M_A with its standard error over ten fresh seeds (equal within two standard errors). Three communities do not give a statistical generalisation.

	Selected design			Installed power		EV passenger change	
Community	M_R	M_A	Same	Difference (kW)	Form	Difference (SE)	Direction
Khayelitsha	n03-01-01_p50	n05-06-02_p50	No	+400	Capacity and hubs	−0.119 (0.004)	Lower
Bellville	n01-03-00_p50	n00-04-00_p50	No	0	Hubs	−0.035 (0.010)	Lower
Wynberg	n01-01-00_p50	n02-00-00_p50	No	0	Hubs	−0.103 (0.006)	Lower

The replay model evaluates all designs of a community in 2.2 to 67.5 s in one process (Table 20). One agent-based run of 28 days takes 0.5 to 3.4 min, and the full agent-based chain of a community (warm-up, a TPE search of 36 to 39 trials, the validation, the fresh-seed evaluation, the sensitivity cases and, in Khayelitsha, the bound check) needs 2.0 to 13.6 h of run time, of which the search is 0.4 to 2.2 h. The agent-based analysis therefore costs about three orders of magnitude more computation than the replay analysis.

Table 20: Computational cost of each phase in each community. Each design evaluation of M_R and each engine run of M_A uses one process. Runs: the designs of M_R or the engine runs of M_A, with the paired ICE runs. Run time: the sum of the wall times of the runs. Elapsed: the wall time of the phase, with two runs at the same time; the search trials record only their run time. Memory: the peak resident memory of one run.

Community	Phase	Runs	Run time (h)	Mean run (min)	Elapsed (h)	Memory (GB)
Khayelitsha	M_R enumeration	2,645	0.019	–	0.019	–
	M_A warm-up (49 days)	1	0.05	3.0	0.05	1.8
	M_A TPE search	39	2.20	3.4	–	–
	M_A validation (selection seeds)	105	5.40	3.1	2.04	2.8
	M_A fresh seeds	50	2.01	2.4	1.15	2.8
	M_A sensitivity	70	3.30	2.8	1.89	3.8
	M_A bound check	15	0.68	2.7	0.40	2.8
Bellville	M_R enumeration	1,169	0.011	–	0.011	–
	M_A warm-up (63 days)	1	0.07	4.4	0.07	0.9
	M_A TPE search	38	2.10	3.3	–	–
	M_A validation (selection seeds)	50	2.17	2.6	1.17	2.3
	M_A fresh seeds	40	1.83	2.7	1.01	2.3
	M_A sensitivity	70	3.59	3.1	1.97	3.3
Wynberg	M_R enumeration	349	0.001	–	0.001	–
	M_A warm-up (91 days)	1	0.02	1.3	0.02	0.5
	M_A TPE search	36	0.43	0.7	–	–
	M_A validation (selection seeds)	45	0.47	0.6	0.18	1.1
	M_A fresh seeds	50	0.44	0.5	0.26	1.1
	M_A sensitivity	70	0.68	0.6	0.39	1.4

The cost model turns the passenger difference into a decision (Table 21). In Khayelitsha the agent-based design costs R2.0 million a year more than the replay design, 61% of it demand charges, and its EV agents carry 147,000 more passengers a year. It pays for itself at a fare of R13.7 a trip (R11.4 to R17.7 over the cases); the 1,000 kW design of the 99% tolerance would need R19.8. The City of Cape Town sampled fares of R14 to R20 for comparable routes in 2022 [59]. In Bellville and Wynberg the agent-based design has the same installed power but costs R40,000 a year less, because its connectors are together at one site, and it carries more passengers; it dominates the replay design in every case. In Wynberg, the 150 kW design of the 99% tolerance would pay for its additional connector at R7.4 a trip.

Table 21: Annual cost and break-even fare of the selected designs (rand of December 2025, without VAT). Connectors: the installed connector cost, annualised over ten years at a discount rate of 0. Demand: the City of Cape Town demand and network capacity charges of the sites. EV passengers: the passengers that the EV agents carry each day (ten fresh seeds). Extra: against $\mathbf{x}_R^\star$, with 250 operating days a year. Break-even fare: the fare at which the extra fare revenue pays the extra cost, in the base case and its range over the discount rates of 4, 8 and 12% and 300 operating days. Dominates: the design costs less and carries more passengers. $\tau = 0.99$: the M_A design of the 99% tolerance where it differs from $\mathbf{x}_A^\star$.

			Annual cost (R thousand)			EV passengers	Extra cost	Extra passengers	Break-even fare (R)
Community	Design	Installed (kW)	Connectors	Demand	Total	a day	(R thousand a year)	a year	[range]
Khayelitsha	$\mathbf{x}_R^\star$	250	665	845	1,510	4,044	–	–	–
	$\mathbf{x}_A^\star$	650	1,444	2,080	3,524	4,633	2,014	147,404	13.66 [11.39–17.73]
	$\tau = 0.99$	1,000	2,046	3,191	5,237	4,799	3,726	188,671	19.75 [16.46–25.39]
Bellville	$\mathbf{x}_R^\star$	200	508	676	1,184	3,016	–	–	–
	$\mathbf{x}_A^\star$	200	469	676	1,145	3,130	−40	28,568	Dominates
Wynberg	$\mathbf{x}_R^\star$	100	313	338	652	932	–	–	–
	$\mathbf{x}_A^\star$	100	274	338	612	1,062	−40	32,741	Dominates
	$\tau = 0.99$	150	431	507	938	1,086	286	38,675	7.40 [6.17–9.73]

6. Discussion

6.1. When does driver agency matter?

The models select different designs in all three communities, but the size and the form of the difference depend on three conditions.

First, the EVs must depend on charging at the hubs during the day. One night of home charging does not cover the daily energy need of 94%–97% of the replayed vehicle-days, so the time that an EV spends at a charger is potentially taken from passenger service when evaluated under M_A. The difference grows with the energy need. With 15% higher consumption, the cost of $\mathbf{x}_R^\star$ (Section 4.7.3) rises from 11.9, 3.5 and 10.3 percentage points in the base case to 15.9, 3.9 and 15.1 points in Khayelitsha, Bellville and Wynberg; with 15% lower consumption, the Bellville difference disappears (+0.1 points, SE 1.2; Figure 7b).

Second, how busy the connectors are decides the form of the difference. Waiting in the queue for a free connector is the main way in which charging costs passengers in M_A, and the more $\mathbf{x}_A^\star$ shortens this wait relative to $\mathbf{x}_R^\star$, the more passengers the EVs keep (Section 5.4). In Khayelitsha, the EVs draw so much energy from the connectors of $\mathbf{x}_R^\star$ that every connector would have to charge at full power for 15.2 h a day to supply it. The connectors are therefore occupied for most of the day, only additional connectors can shorten the queue, and M_A adds 400 kW. In Bellville and Wynberg, the connectors have spare time, and M_A instead puts all of them at one hub at the same installed power. The EVs then share one queue for several connectors instead of a separate queue at each hub. Such a shared queue moves faster than separate queues with the same total number of connectors, which is known as the pooling effect in queueing theory [76, 77]. The time that the EVs spend in the queue falls by 75% in Bellville and by 58% in Wynberg. The replay model rewards the opposite. A replayed vehicle can charge only at a hub where it already stops, so M_R spreads the connectors over the hubs at which the observed vehicles stop, and it rates the single-hub design as worse because the vehicles that do not stop at that hub cannot use its connectors (Table 14).

Third, the difference is smallest where the observed vehicles already stop at the hubs. Bellville has the fewest replayed vehicle-days without a stop of 10 min or more within 100 m of a candidate hub (8.1%, against 28.5% in Khayelitsha and 33.3% in Wynberg; Table 9), and it has the smallest cost of $\mathbf{x}_R^\star$. The screening does not anticipate this. The binding of hub charging ($B = V_2 - V_1$; Table 6) is largest in Bellville (49.7 pp), but it is large because its vehicles already stop at the hubs, which is the case in which replay is most adequate.

The number of ICE competitors, by contrast, has little effect: halving the scale factor k, which sets the size of the ICE population and of the passenger demand, or increasing it by half changes the cost of $\mathbf{x}_R^\star$ by at most 2.1 percentage points (Figure 7b). Competition does not set how many passengers the EVs lose; it sets where those passengers go as further discussed in Section 6.3.

A stronger replay baseline would let a vehicle divert to the nearest charger when its battery is low. Such a rule would remove part of the stranding of M_R at a fraction of the computational cost. Without a model of the queue at the charger, however, it would not show the passengers that EVs lose while they wait, and by sending each vehicle to its nearest hub it would not find the single-hub designs of Bellville and Wynberg. A queue-aware diversion rule would close part of the gap, but it would then need the same calibration of when and where drivers charge as the agent-based model.

6.2. Two models, two verdicts

The cross-evaluation shows that the choice of model is not neutral. In Bellville and Wynberg, each model's design is worse in the other model: the agent-based design strands more vehicle-days in M_R, and the replay design carries fewer EV passengers in M_A (Table 14). Had both models scored the designs on one shared objective, this disagreement would not have been visible. It appears because each model selects its design using the outcomes that it can represent (stranding in M_R, EV passengers in M_A), and the design is then evaluated in the other model. The model therefore determines which outcomes the planner can see [8].

The models also differ in how electrification fails. In M_R, an EV fails by running out of energy: under $\mathbf{x}_R^\star$, it strands on 57.0%, 22.6% and 49.0% of the vehicle-days in Khayelitsha, Bellville and Wynberg. In the base case of M_A, no EV strands. Instead, the EVs carry fewer passengers, because their drivers spend time charging and the ICE competitors carry the passengers they miss. The replay model therefore overstates the energy infeasibility of electrification and cannot show its cost in passenger service.

Neither model predicts the outcome of a pilot on its own. The agent-based drivers do not strand because the charging rules give them a reasonable degree of foresight: they check the energy of a route and of the drive to a station before they start it (Appendix C.7.5). M_R assumes that drivers do not adapt, and M_A that they adapt competently from the first day. Drivers in an early pilot will probably lie between the two, so the models are better read as bounds than as alternatives.

6.3. Implications for electrification pilots

Four implications follow for the planning of pilots. First, where the connectors go matters as much as how many are installed. In Bellville and Wynberg, putting all connectors at one hub keeps 3.5 and 10.3 percentage points more EV passengers than spreading the same installed power over two hubs, and it costs about R40,000 a year less, because one larger installation is cheaper per connector than two smaller ones (Table 21).

Second, additional capacity is a close economic decision. In Khayelitsha, the extra 400 kW of the agent-based design pays for itself if each additional EV passenger pays a fare of at least R13.7, just below the fares of R14 to R20 sampled on comparable routes in 2022 [59]. The City of Cape Town's demand and network-capacity charges make up 61% of the extra cost, so the electricity tariff and any limit on the peak site load affect the decision as much as the cost of the connectors. The selected capacity also depends on how close to the service of the best design the selection rule requires a design to be: from 90% to 99% of it, the Khayelitsha design grows from 374 to 1,000 kW (Table 12). Above about 600 kW of peak site load, each further connector gains little. The robust result is the direction of the change, not its size in kilowatts.

Third, a pilot should monitor the passengers and revenue of the EV operators, not only the service of the system. In Khayelitsha and Wynberg, the ICE competitors take 85% to 98% of the passengers that the EVs lose, and the share of passenger requests that expire unserved changes by at most 0.04 percentage points. Measures of the service of the whole system would therefore show no effect, while the time-adjusted passenger loss ℓ of the EVs remains 5.6% in Khayelitsha and 16.4% in Wynberg even under $\mathbf{x}_A^\star$. This loss falls on the early adopters and risks detering adoption. Careful consideration of the fare and of the revenue of the EV operators is therefore needed for a pilot to succeed.

Fourth, the two models suit different stages of planning. Replay evaluates every design of a community in at most about a minute and suits the screening of communities and the generation of candidate designs. Its own outputs show when the agent-based model is needed: when many vehicle-days strand even with unlimited connectors, when many days have no stop near a candidate hub, and when the connectors would have to charge at full power for most of the day. At pilot scale, the run time of the agent-based analysis (2 to 14 h for a community) is not the main cost; the calibration of behaviour and demand is.

These findings support evaluating candidate designs with both energy-feasibility and passenger-service measures. They do not establish which sites are technically or commercially feasible. The candidate hubs, the design space and the charging assumptions remain a model of a possible pilot.

6.4. Limitations

Movement representation. Calibration reproduces the median daily distance and moving time, and three remaining differences have a known direction. The simulated vehicles spend 1.5 to 2.5 h longer at home overnight than the observed vehicles, which gives the EVs 5 to 8 kWh more home charging a day (6% to 15% of the median daily energy need), so M_A probably understates the need for hub charging and the

passenger losses. The simulated vehicles vary less from day to day, so the extreme days that lengthen the queues are under-represented. The inferred network misses 18.6% to 41.0% of the observed driving, most in Wynberg, whose drive-to-charger time and remaining loss are therefore the least certain. The decomposition of Table 15 shows that the simplified movement does not explain the difference between the selected designs; in Khayelitsha it even works against it.

Behavioural assumptions. Drivers queue for a connector in order of arrival and do not know the queue lengths at other hubs, which probably overstates the queue time compared with real drivers who could share this information. Drivers pay no price for charging, and every vehicle with a home charges there at 3.7 kW. The passenger demand is not observed; it is scaled to the full taxi population by k, and the Bellville value exceeds the upper bound from the operating licences in the provincial route register. The sensitivity cases from $0.5k$ to $1.5k$ change the cost of $\mathbf{x}_R^\star$ by at most 2.1 points, so the comparison is robust to this value, although the absolute passenger numbers are not.

Comparison and cost. The models measure service differently (stranded vehicle-days in M_R, EV passengers in M_A), so the behaviour component of the decomposition also contains this change of measure and is not a causal estimate of behaviour alone. The cost model uses US connector costs, which are probably high for South Africa, and fares that are not published for the studied routes, so its result is a break-even fare, not a business case.

Generalisation. The screening selects communities partly on their replay outcomes, so the selected communities are those in which replay performs best; this should narrow rather than widen the difference. Three communities in one city and two December weeks cannot establish generalisability across South African minibus-taxi operations or across seasons. As Collett and Hirmer [78] argue, generalisable planning models for paratransit electrification require more systematic data collection.

6.5. Future work

Four extensions follow from the results. A queue-aware replay with diversion to a charger would show how much of the agent-based result a cheaper model can recover. Reevaluation with candidate sites chosen from where vehicles dwell and operate, rather than the main hubs only, would address the drive-to-charger cost. Constraints and tariffs of the distribution network, together with managed charging, would address the demand charges that dominate the cost of additional capacity. Finally, observations of when and where drivers charge in the first real-world pilots would calibrate the charging behaviour and show where real drivers lie between the two models. A replication with more communities, a larger fleet and other seasons would test the generalisability of the findings.

7. Conclusion

Representing driver adaptation changed charging design selection in all three communities, but the form of the change depended on local operating patterns. In Khayelitsha, the agent-based model selected substantially more installed power; in Bellville and Wynberg, it selected a different connector allocation at the same installed power. Across all three communities, the replay-selected designs carried 3.5 to 11.9 percentage points fewer EV passengers when evaluated in the agent-based model on seeds that the selection did not use, and most of these passengers went to internal-combustion competitors. The additional Khayelitsha capacity pays for itself at a fare of about R14 a trip, and in Bellville and Wynberg the agent-based design is also the cheaper design. These results suggest that replay-based analysis remains useful for evaluating electrification feasibility under observed mobility, but that it may not identify the charging design that best supports passenger service when drivers can adapt. Agent-based simulation can represent these additional operational effects, provided its demand assumptions, behavioural calibration and movement representation are assessed explicitly.

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Appendix A. Validation of the vehicle communities

Appendix A.1. Similarity graph and partition

Vehicle communities are detected with a multi-view similarity graph. Each vehicle is described by four views of non-negative feature weights, which are computed from the stop events and edges inside the study area on valid weekdays: the dwell hours at each non-home node, the vehicle-days on each network edge, the dwell hours at each formal node, and the vehicle-days with each pair of consecutive anchor dwells (origin-destination pairs). The features of each view are weighted with term frequency-inverse document frequency (TF-IDF) [**?**]:

$$\tilde{x}_{vf}^{(m)} = \ln\left(1 + x_{vf}^{(m)}\right)\left(\ln\frac{1+n}{1+\mathrm{df}_f} + 1\right), \tag{A.1}$$

where $x_{vf}^{(m)}$ is the weight of feature f for vehicle v in view m, n is the number of vehicles, and df_f is the number of vehicles that use feature f. Each weight vector is subsequently normalised to unit length, such that a hub used by the whole fleet contributes less to the similarity of two vehicles than a stop shared by only a few. The combined similarity $S_{uv} \in [0, 1]$ of vehicles u and v is the mean cosine similarity over the views that both vehicles have. A similarity graph joins each vehicle to its 15 most similar vehicles, and the Leiden algorithm [**?**] partitions the graph by maximising the modularity with a resolution parameter γ [**? ?**]:

$$Q_\gamma = \frac{1}{2W} \sum_{u,v} \left[A_{uv} - \gamma \frac{k_u k_v}{2W} \right] \mathbf{1} \left[c_u = c_v \right], \tag{A.2}$$

where A_{uv} is the edge weight between vehicles u and v, k_u is the strength of vehicle u, W is the total edge weight, c_u is the community of u, and $\mathbf{1}[\cdot]$ is the indicator function. A larger γ produces smaller communities.

For each resolution $\gamma \in \{0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 2.5, 3\}$, the Leiden algorithm is run with 50 seeds, and the seed partitions are combined into a consensus partition [**?**]. A resolution is eligible when its median seed partition contains at least two communities of five or more vehicles, and when its consensus partition retains at least 90% of corridor use within one community, where a corridor is an undirected pair of consecutive anchor dwells that recurs on at least three vehicle-days and two dates. The latter condition encodes the operational meaning of a community, namely vehicles that share corridors in a common operating area. Of the eligible resolutions, the most stable resolution, as measured by the mean pairwise adjusted Rand index (ARI) [**?**] of its seed partitions, is selected. Vehicles in communities of fewer than five vehicles are not assigned to a community. The existence of the communities is tested against a null model of randomised fleets [**?**], and their seed stability, temporal stability and agreement across views are given in Table A.22.

Appendix A.2. Evidence that the communities exist

The observed modularity of the accepted partition is 0.698, against 0.495 (standard deviation 0.015) for 100 randomised fleets with the same row and column totals, which gives a z-score of 13.1 and a p-value of 0.0099, the smallest p-value that 100 randomised fleets can give (Table A.22). The partitions of the 50 seeds agree with a median pairwise ARI of 0.891. A partition from the early dates against one from the late dates, over the 315 vehicles in both halves, gives an ARI of 0.620 and a normalised mutual information (NMI) of 0.639, so the communities persist over the study period but some vehicles change community. Each single view agrees with the combined partition with an ARI of 0.486 (edges) to 0.775 (ranks). The accepted partition keeps 91.8% of the use of the 498 recurring corridors within one community.

Table A.22: Evidence tests of the accepted community partition. Null model: 100 randomised fleets by curveball trades. Seed stability: the median pairwise ARI of the partitions of 50 seeds. Temporal stability: the partition of the early dates against that of the late dates. View agreement: the partition of each view against the combined partition.

Test	**Value**
Modularity, observed	0.698
Modularity, 100 randomised fleets, mean (SD)	0.495 (0.015)
z-score and p-value of the observed modularity	13.1 and 0.0099
Seed stability, median pairwise ARI	0.891
Temporal stability, ARI and NMI (315 vehicles in both halves)	0.620 and 0.639
View agreement, nodes, ARI (NMI)	0.746 (0.777)
View agreement, edges, ARI (NMI)	0.486 (0.579)
View agreement, ranks, ARI (NMI)	0.775 (0.843)
View agreement, OD pairs, ARI (NMI)	0.632 (0.680)
Corridor use in one community (498 recurring corridors)	0.918

Appendix B. Network construction and coverage

Appendix B.1. Stop events and homes

Stop events are extracted from the GPS trajectories with an anchored-dwell algorithm that tolerates isolated positioning errors. A candidate stop is anchored at the median position of its first three inlier points, and a later point is an inlier when it lies within 30 m of the anchor and the pairwise diameter of the

inliers remains at or below 60 m. One isolated outlier is permitted when the subsequent point returns to the candidate, whereas a second consecutive outlier, or a reporting gap of more than 300 s, ends the candidate. A candidate becomes a stop event when it contains at least three inlier points, has an outlier share of at most 10%, and lasts at least 180 s, a value that was selected from a sweep of 30 to 600 s. At the sampling interval of about 80 s, three inlier points span at least about 160 s. Consequently, stops shorter than about 2 to 3 minutes, which include most of the boarding and alighting stops along a route, cannot be observed in the data. The location of a stop event is the median of its inlier positions.

The home base of each vehicle is identified with a boundary-point rule. The first and last points of each valid vehicle-day are collected, the boundary point with the most other boundary points within 300 m, the home radius, is selected as the candidate, and the home location is the median of the boundary points within the home radius of the candidate. A vehicle has a home when both the first and the last points of at least 75% of its valid vehicle-days lie within the home radius of this location; otherwise, its most frequent first point is used as its start location. The sensitivity of the rule was assessed for radii from 150 to 1,000 m and minimum shares of 0.6, 0.75 and 0.9.

Appendix B.2. Nodes, edges and review

A single operational network is constructed for the whole study area and is used by all communities. Rather than representing the full road network, the network represents the locations at which taxis stop in an operationally important way, the fixed infrastructure that the fleet uses, and the links between these locations, such that nodes are derived from evidence of stopping and edges from evidence of driving. The eligible stop events (inside the study area, on valid weekday vehicle-days, and not at a home unless that home is at infrastructure) are counted in two square grids of 100 m cells with offsets of 0 and 50 m, so that a rank on a cell boundary is not split into two weakly supported cells. A cell is selected when it contains stop events from at least five vehicle-days and three distinct vehicles. This support rule replaces the classifier of an earlier stop-network method and achieved a higher precision and F1 score than the classifier on held-out labelled cells from Gauteng. The selected cells are merged into support nodes by connected complete-linkage clustering with a maximum diameter of 200 m, which prevents long spatial chains, and each stop event is associated with one node only.

Formal nodes are obtained from the City of Cape Town minibus-taxi route termini, the taxi ranks and railway stations in OpenStreetMap (OSM), and one large rank that these sources omit, namely the Site C rank in Khayelitsha, which was confirmed on satellite imagery (Section 3.3). Formal nodes within 30 m of each other are merged into one node, a formal node is retained only when the fleet stops there, and a support node within the snap distance of 100 m of a formal node joins it and adopts its location. Home bases that are not at infrastructure become home nodes, which vehicles can reach but cannot traverse.

Since a moving vehicle covers a median distance of 435 m between consecutive GPS points, a vehicle can pass through several nodes between two stop events without a detected stop. A pass is therefore recorded when the straight line between two consecutive GPS points comes within 100 m of a node, with the pass time and cumulative distance interpolated at the point of closest approach. Passes split the movement between two stop events into hops, but they do not count towards the support of a node, so that nodes remain defined by where taxis stop. The network contains three kinds of directed edges: observed edges, which join the two nodes of a hop and are retained when at least two vehicle-days use them; derived reverse edges for observed edges without an observed counterpart; and close-node links in both directions between nodes within 300 m of each other, which allow short movements between nearby nodes without long detours. All edge lengths are shortest road paths on the drivable OSM network, such that one-way streets are respected.

Rather than relying solely on automatic rules, the network is subjected to a single manual review of 211 items, which comprise candidate cells with high vehicle-day support but few vehicles, edges with implausible ratios of road length to straight-line length or implausible speeds, and conflicts between support nodes and formal nodes. Each item was reviewed in an interactive map that plays back the relevant vehicle-days, and 2 of 30 candidate cells, 153 of 168 flagged edges and all 13 conflicts were accepted. The resulting network comprises 726 nodes, of which 18 are ranks, and 98.9% of the nodes lie in the largest strongly connected component. The nodes capture 76% of the eligible stop events and 72% of their dwell time, and 66.5% of the distance driven between stop events runs on network edges.

Appendix B.3. Route library and route sets

Since the intermediate nodes of a taxi vary from trip to trip between the same terminals, the observed trips are compressed into representative routes. For each ordered terminal pair, the distinct node sequences are processed in order of decreasing support, and a sequence joins the most similar existing variant when the Jaccard similarity of their node sets is at least 0.5; otherwise, it founds a new variant. A variant qualifies as a route when its trips occur on at least three vehicle-days and two dates. The representative of a qualifying variant is its medoid, i.e., the observed trip with the highest mean similarity to the other trips of the variant, and the stops of the route are the terminals and stop nodes of the medoid, together with each pass that at least half of the variant's trips visit. This majority rule removes the noise of parallel roads while retaining the nodes at which taxis consistently search for passengers. All route nodes are possible stops, and whether a driver stops at a node is a behavioural decision in M_A (Section 4.4).

The route library is built from the trips of the whole fleet, because a route is a property of the operation and more vehicles provide more support. Each route receives a reverse counterpart, which is either the most similar qualified route in the opposite direction or a derived route with the reversed node sequence. Additionally, each loop is split at its turning node (the node farthest from the origin) into an outward and a return half, so that a taxi can turn at the turning node and start a new run there. The resulting library comprises 586 qualified routes and 503 derived reverse routes.

The eligible routes of a driver are the smallest set of its routes, ordered by their share of its trip distance on qualified routes, that covers 80% of that distance, together with the reverse counterparts of these routes and the two halves of each eligible loop. In the selected communities, routes shorter than 5 km in the network are excluded, since a route whose node sequence retains only a few nodes can be much shorter in the network than its observed trips and would cause an agent to drive too little. A vehicle with at least one eligible route qualifies for the models: both models electrify it, and it becomes an EV agent in M_A. The routes carry no time of day, and the direction and timing of demand are instead handled by the demand model (Section 4.5). Finally, a route is treated as a single unit of operation in M_A: once a driver has started a route, it cannot charge at a rank along that route. Consequently, an EV does not select a route that it cannot complete from a full battery (Section 4.4).

Appendix C. Agent-based model specification

This appendix describes M_A as configured in this study, with the seven elements of the Overview, Design concepts and Details (ODD) protocol in its second update [71]. Since the drivers' decisions are the subject of the comparison, the design concepts also answer the questions of the ODD+D extension for human decision-making [72]. The model is the discrete-event engine of Sello and Booysen [46] with the changes of Section 4.4 and Appendix C.8. Table C.25 gives the values of the behavioural parameters.

Appendix C.1. Purpose and patterns

The purpose of M_A is to predict the outcomes of a charging design for an electrified minibus-taxi community when the drivers can adapt to it: when they work, which routes they drive, how long they wait for passengers, when they idle and when and where they charge. The model therefore generates the movement of the vehicles and the passenger service, which M_R takes as given or cannot represent (Table 4).

The model is evaluated against these patterns: the distributions of the daily distance, the daily moving time and the day-to-day variation of the daily distance of the observed vehicles (the calibration targets of Table 8); the hourly share of moving vehicles, as a realism check; a served share of at least 95% of the requests; and a seat occupancy on route runs of up to 0.6. The fidelity check of Section 4.6.2 also compares the simulated movement with the observed movement in the quantities that the replay rules use.

Appendix C.2. Entities, state variables and scales

Table C.23 lists the entities of the model and their state variables. A driver is in one of eight states: at home, on a route, idling, heading home, seeking a charger, waiting to charge, charging or stranded. The last four states apply only to EVs.

Time is continuous, in seconds from the start of the run, and one day is 86,400 s. Every simulated day is the same typical term weekday: all days have the same shift distributions and the same demand. Space

Table C.23: Entities of M_A and their state variables.

Entity	State variables
Node	Identifier; a node exists only as an end of an edge and can be a rank
Rank	A first-in, first-out loading queue of the on-route taxis at a formal rank
Edge	A directed link between two nodes, with a fixed road length and a travel-time model (Appendix C.7.2)
Route	An ordered list of stops; consecutive stops are connected on the network
Driver	Home node, idle locations, eligible routes, shift distribution, vehicle, policies, memory, state, location, passengers on board, and the passengers and passenger distance of the current day
Vehicle	ICE: 15 seats. EV: also the usable battery capacity, the energy consumption rate and the energy level
Request	Origin, destination, request time, boarding time, arrival time and expiry time
Memory	The streaming statistics of the route runs and the charging visits of a driver in each 30-minute interval of the day
Charging station	Connectors, connector power and a first-in, first-out queue (EV runs only)

is the operational network of the community, with 561 to 598 nodes, including the homes of the drivers, and directed edges between them. The model has no positions between nodes. A run has a warm-up, 14 settling days and 14 evaluation days (Appendix C.5).

Appendix C.3. Process overview and scheduling

The engine is a discrete-event simulation in continuous time [79]. A priority queue holds the scheduled events, and the earliest event is processed first. Events with the same time are processed in the order in which they were scheduled. Table C.24 lists the events.

Table C.24: Events of M_A and their main effects.

Event	Main effect
Day start	Expires the requests that still wait from the previous day, generates the requests of the day, restores each stranded driver at home with a full battery and schedules the next day start 24 h later
Trip request	The passenger boards a compatible taxi that is stopped at the origin node, or otherwise waits there
Trip expiry	The request expires if it still waits
Driver departure	Off a route: applies home charging and starts the decision procedure. On a route: the driver leaves the stop for the next stop
Driver arrival	Applies the energy and time of the travel and then acts on the state of the driver, for example the stop procedure at a stop
Driver evaluation	Evaluates a driver that waits at a stop, idles, waits to charge or charges
Charging stopped	Closes the charging session and gives the connector to the next driver in the queue

A driver that is not on a route decides in this order. First, it returns home if its shift has ended; an EV that would arrive home below its reserve charges first. Second, an EV that is not full consults its charging policy. Third, the driver chooses between its routes and idling (Appendix C.7.3). An EV whose energy does not cover the chosen route, the drive from its last stop to the nearest station and its reserve charges first. A chosen route sends the driver to the first stop of the route.

At each stop of a route, the passengers for the stop alight, and the route ends at its final stop. An empty driver whose shift has ended goes home, and a full taxi departs. Otherwise, the expired requests are removed, the compatible requests board, the driver joins the loading queue if the node is a rank, and the stop decision of Section 4.4 determines whether the driver waits or departs.

Appendix C.4. Design concepts

Basic principles and theoretical background. The drivers are boundedly rational: they choose among their eligible routes and idling with a multinomial logit model [73] that uses remembered and currently observed demand. The model follows minibus-taxi operation rather than demand-responsive dispatch: taxis load at ranks on a first-in, first-out basis and depart when sufficiently full, drive fixed routes between terminals, and wait at stops according to their remembered demand. No operator assigns requests to vehicles. The behavioural parameters are calibrated to the duty cycles of the observed vehicles (Section 4.6).

Emergence. The movement of the vehicles, their daily distance and moving time, the site load, the charger queues, the passenger waits and the distribution of passengers between the EV agents and the ICE competitors emerge from the decisions of the drivers and their competition for requests. None of these quantities is an input.

Adaptation and individual decision-making. A driver decides when to return home, which route to drive or whether to idle, and how long to wait at each stop. An EV driver also decides when to charge. Each decision is a stochastic choice whose probabilities depend on the remembered demand of the current interval of the day, the current demand at the node, the occupancy of the vehicle and, for an EV, its energy level.

Objectives. The utility of a route increases with the waiting requests that it can carry, the remembered passengers and passenger distance of its runs, and decreases with the remembered run distance and the competing drivers at the node (Appendix C.7.3). The utility of idling increases when the driver remembers the current interval as quiet. The drivers do not optimise an income over the day.

Learning. The memory of each driver records its route runs and charging visits as streaming means and variances in 30-minute intervals of the day, with no decay. The memory persists over days, so a driver learns where and when its routes carry passengers. In a cold run, the temperature of the choice model decreases from 1 to 0.1 over ten days, such that the drivers first try all their routes. The EV drivers learn their charging behaviour during the 14 settling days.

Prediction. A driver predicts the passengers of a route from its memory of the current interval. With the last-run rule, a driver predicts the end time of a route from the remembered duration or, without a memory, from the shortest-path durations between its stops. An EV predicts the energy of a route, of the drive from the last stop to the nearest station and of the drive home from the shortest static paths.

Sensing. A driver senses the waiting requests at its current node that each of its routes can carry, the other on-route drivers at the node that drive the same route, the passengers on board, its own energy level and its memory. It does not sense the demand or the queues at other nodes.

Interaction. The drivers interact indirectly through competition: a request boards the first compatible taxi at its origin, in the loading queue at a rank or by arrival time at another node. EV drivers also interact through the connectors and the first-in, first-out queues at the charging stations.

Stochasticity. The engine uses its own random number generator, and two runs with the same configuration and seed are identical. The working days and shift times, the travel times, the request counts and times, the route choices, the stop decisions, the intervals between evaluations and the idle locations are random draws. A feature without information, for example a route without a memory in the current interval, is a uniform random number. The engine seed of a run is 101 plus the seed offset of the run. The synthetic population is a separate seeded draw, so all runs of a community have the same drivers.

Collectives. The model has no collectives. A community, a rank queue and a charger queue do not make decisions.

Heterogeneity. Each agent has its own home, eligible routes, idle locations and shift distribution, which come from its observed vehicle. A synthetic copy has the routes, idle locations and shift pattern of its agent, a nearby home and a shifted shift. All drivers of a community have the same behavioural parameters (Table C.25).

Observation. The engine logs the route runs, the requests, the driver states and the charging sessions of the evaluation days. The measures of a design (Table 5) and of the calibration (Table 8) are computed from these logs. A simulated agent-day is one day from 00:00 to 24:00.

Appendix C.5. Initialisation

The build of a community creates the drivers from the EV agents and their ICE competitors (Appendix C.7.1), with a fixed seed. A cold run starts with empty memories and runs a warm-up period followed by 14 evaluation days. The warm-up length of each community, between 49 and 91 days, is the first day after which the seven-day means of the served passengers, their share of the requests, the occupancy and the route-choice shares remain within a tolerance of their long-run means on three seeds, rounded up to whole weeks. To reduce the computational cost, the warm-up is run once, and the resulting driver memories are saved as a warm start. Each subsequent run loads these memories, disables exploration, and runs 14 settling days followed by 14 evaluation days. The warm-up lengths are 49, 63 and 91 days in Khayelitsha, Bellville and Wynberg, respectively. In the experiments, the EVs start the settling days with a full battery, and the seed offsets of the five validation runs give the engine seeds 102 to 106. The cross-evaluation uses the fresh engine seeds 201 to 210, and the sensitivity cases use 201 to 205; no fresh seed is used to select a design.

Appendix C.6. Input data

The model uses no external time series during a run. All days use the same derived inputs: the network and its travel-time models (Appendix C.7.2), the eligible routes, homes and idle locations of the agents (Section 4.2.2), their shift distributions (Appendix C.7.1), the origin–destination demand and its hourly profiles (Appendix C.7.6), the vehicle and charging values (Section 4.1.1) and, in the experiments, the charging design.

Appendix C.7. Submodels

Table C.25: Behavioural parameters of M_A in each community. The first nine parameters are calibrated (Appendix D). Khayelitsha started its search from the calibrated values of Wynberg, and no validated trial fitted its duty cycles better, so it keeps the Wynberg values (Table D.27); its model differs from that of Wynberg in the travel-time scale, the last-run rule, the demand and the observed shifts. The other parameters are fixed in all communities.

Parameter	Symbol	Khayelitsha	Bellville	Wynberg
Idle bias		0.207	−0.793	0.207
Idle demand weight		1.18	2.61	1.18
Stop bias	b	0.181	3.12	0.181
Maximum stop duration (s)		301	674	301
First-stop bias	b_1	−1.43	−2.41	−1.43
First-stop demand weight	$w_{d,1}$	16.9	26.6	16.9
Maximum first-stop duration (s)		3,121	3,932	3,121
Shift spread		0.466	0.189	0.466
Shift end offset (s)		−982	−5,094	−982
Stop demand and occupancy weights	w_d, w_o	2.0 and 4.0		
Route weights: demand, competition, distance		0.10, 0.25 and 0.10		
Route weights: passengers, passenger distance		0.30 and 0.25		
Idle income weight		0.05		
Final temperature and exploration time	τ	0.1 and 10 days (0 in warm-started runs)		
Mean interval between stop evaluations (s)		60, exponentially distributed		
Memory interval (min)		30		
Request expiry (h)		2		
Last-run rule		On	Off	Off
Travel-time excess factors c_w		1.31–1.50	–	–

Appendix C.7.1. Population and shifts

The drivers of a community comprise its EV agents and their ICE competitors, which are synthetic ICE copies of the EV agents. Each agent keeps its home node, its eligible routes and its idle locations, which are the terminals of its routes. Since the observed vehicles are only a sample of the taxis that operate from a hub, each agent receives $\lfloor k-1 \rfloor$ or $\lceil k-1 \rceil$ ICE copies from a seeded Bernoulli draw, such that the population is on average k times the number of agents, where k is the scale factor of Section 4.5. A copy adopts the routes, idle locations and shift pattern of its agent, its home is drawn from the five network nodes nearest to

the agent's home, and the start and end of its shift are moved by a common normal draw with a standard deviation of 30 min. This construction rests on the assumption that the EV agents are representative of the taxis that load at the hub of the community, both in the routes that they drive and in their working hours. The resulting populations comprise 1,358, 1,673 and 686 drivers in Khayelitsha, Bellville and Wynberg, respectively, of which the EV agents form only 2.6% to 6.6% (Table 7).

Each agent has a typical-weekday shift, of which the start and end times are normally distributed with the mean and standard deviation of the first and last movements on its calendar days in the calibration period, and each day is a working day with the observed probability of activity. Two calibrated parameters, the shift spread and the shift end offset, scale the standard deviations and move the mean end time, respectively. The offset is necessary because the observed end of a shift is the arrival at home, whereas the simulated end is the time at which the driver starts to return home. In Khayelitsha, a last-run rule additionally prevents a driver from starting a route that it expects to end after its shift end, because too many agents otherwise remained active late at night.

Appendix C.7.2. Travel time

The travel time of an edge ℓ that a driver enters in time window w is modelled as follows:

$$T_\ell = \frac{s_\ell}{80 \text{ km/h}} + X_{\ell w}, \qquad X_{\ell w} = \begin{cases} 0, & \text{with probability } \pi_{\ell w}, \\ c_w Y, \quad \ln Y \sim \mathcal{N}\left(\mu_{\ell w}, \sigma_{\ell w}^2\right), & \text{otherwise}, \end{cases} \tag{C.1}$$

where s_ℓ is the road length of the edge, $\pi_{\ell w}$ is the share of observed hops without excess time, $\mu_{\ell w}$ and $\sigma_{\ell w}$ are the parameters of a log-normal fit of the observed excess times, and c_w is an optional excess factor. The four time windows are the night (19:00 to 06:00), the AM peak (06:00 to 09:00), the midday (09:00 to 15:00) and the PM peak (15:00 to 19:00). The fit uses the hops of the whole fleet on the calibration days, excluding hops slower than 5 km/h, since such hops contain a stand rather than driving and would make the simulated vehicles much slower than the observed vehicles. Edges with fewer than five retained hops receive a fixed excess from the pooled speed of the window, reverse edges copy the model of their observed counterpart, and close-node links use the median speed of the short observed edges. In Khayelitsha, where the simulated agents drove faster than the observed vehicles in every window, the excess is scaled by a factor c_w for each window to match the observed median speed. Drivers travel along shortest paths by road length, and the model does not represent congestion. Only the 18 formal ranks of the network act as engine ranks with loading queues.

Appendix C.7.3. Route choice

A driver that is not on a route first returns home when its shift has ended. Otherwise, an EV driver consults its charging policy, after which the driver chooses between its routes and idling with a multinomial logit model [73]:

$$\Pr(j) = \frac{\exp\left(U_j/\tau(t)\right)}{\sum_i \exp\left(U_i/\tau(t)\right)}, \tag{C.2}$$

where U_j is the utility of route j or of idling, and $\tau(t)$ is a temperature that decreases from 1 to 0.1 during an initial exploration period of ten days, such that drivers initially sample all of their routes. The utility of a route is a weighted sum of normalised features: the waiting requests at the current node that the route can carry, the competing drivers at the node on the same route, and the remembered distance, passengers and passenger distance per run of the route in the current interval of the day. The utility of idling comprises an idle bias, a demand term with an idle demand weight that increases as the driver's remembered demand in the current interval decreases, and a small income term, such that drivers idle more during intervals that they remember as quiet. The route weights are fixed at the values of an earlier calibration of the model, whereas the idle bias and the idle demand weight are calibrated (Section 4.6).

Appendix C.7.4. Boarding and memory

A passenger boards only a taxi that is at the origin node, has a free seat and serves a route on which the destination follows the origin; at a rank, the first compatible taxi in the loading queue is selected. Otherwise, the request waits and expires after 2 h. Passengers do not walk to other nodes or switch to other modes. The memory of each driver records the distance, duration, passengers and passenger distance of each

completed route run, as well as the queue length, waiting time and charging time of each charging visit, as streaming statistics in 30-minute intervals of the day. A driver therefore learns where and when its routes carry passengers.

Appendix C.7.5. Charging and reserve rules

To prevent avoidable stranding, an EV maintains a reserve of $0.2\,C$. It does not select a route that a full battery cannot complete together with the drive from the last stop to the nearest station and the reserve. Moreover, it starts a route only when its energy covers the drive to the first stop, the route, the drive to the nearest station and the reserve, and otherwise charges first, and it charges before returning home when it would otherwise arrive below the reserve. At a station, a driver takes a connector only while it still needs to charge: before its shift end, while it is below its charge target or the energy needed for its cheapest route, or while its policy requests charging; and after its shift end, only while it would otherwise arrive home below the reserve. Home charging at 3.7 kW is applied for the time at home. An EV that nevertheless strands loses the passengers on board and is restored at home with a full battery at the next day start. These rules are corrections of the engine that replace its earlier charging behaviour without a setting to revert to it. Since ICE drivers never charge, the corrections do not change the ICE runs, which was verified by reproducing the logs of a frozen ICE run exactly.

Appendix C.7.6. Passenger demand

A departure is an anchor-to-anchor trip of an agent on the calibration days that is at least 1 km long and leaves the catchment of its origin. The catchment is defined as all nodes within 300 m of the same facility of the Comprehensive Integrated Transport Plan (CITP) of the City of Cape Town or, if no such facility exists, the origin node alone, which removes short movements within a rank complex. Since the stop rates, slow-moving shares and speeds of the observed trips do not distinguish loaded from empty legs at this sampling interval, the direction of demand is taken from survey data. Each departure receives a load factor λ, which equals 1 except for departures in the counter-peak direction of a corridor during the AM peak (06:00 to 09:00) and in the AM-peak direction during the PM peak (15:00 to 19:00). For these departures, λ equals the ratio of counter-peak to peak vehicle trips on the corridor in CITP Table 3-18, which ranges between 0.09 and 0.39, or the mean ratio of 0.21 for corridors without a survey. The peak direction of a corridor is the surveyed direction with more AM-peak trips or, for corridors without a survey, the direction from its residential end, which is identified from OSM land use. Since the agents drive only their eligible routes in M_A, the departures of each agent are moreover redistributed onto its eligible routes in proportion to their load factors with a weight ω_d, such that each agent retains its total load and the supply and demand remain consistent with what the agents drive.

The scale factor is the ratio of the full population of taxis that load at the main hub to the agents:

$$k = \frac{B}{Q\,L}, \qquad L = f \sum_{d \in \mathcal{H}} \lambda_d\, \omega_d, \qquad f = \frac{n_{\text{active}}}{n_{\text{valid}}\,|\mathcal{D}_c|}, \tag{C.3}$$

where B is the daily number of boardings at the CITP facility of the main hub (Section 3.3), $\mathcal{H}$ is the set of agent departures from the hub catchment on valid vehicle-days, and L is the number of full loads of the agents per day. The day factor f converts counts on valid vehicle-days to counts per calibration day, where n_{active} and n_{valid} are the active and valid agent vehicle-days and $|\mathcal{D}_c| = 6$ is the number of calibration days, which corrects for the strong variation of the valid share between dates. This formulation assumes that the routes of the community represent all routes of its hub and that departures in the peak direction and off-peak depart full. The resulting scale factors are given in Table 7. As a plausibility check, k is compared with an upper bound that is derived from the operating licences in the provincial route register. The Bellville value exceeds this bound, which is discussed as a limitation in Section 6.4.

Each departure with load factor λ on a route with node sequence $s_0, s_1, \ldots, s_m$ generates $Q\lambda$ terminal boardings at s_0, which alight at the downstream nodes in proportion to the node weights $w_n = 1 + c_n$, where c_n is the number of agent stop events at node n on the calibration days. The floor of one gives every route node some alightings, since short stops cannot be observed in the data. A further $Q\lambda(1-\alpha)/\alpha$ passengers board and alight between the en-route nodes in proportion to $w_{s_i} w_{s_j}$, such that a share $\alpha = 0.8$ of all

passengers boards at the terminal. The terminal share α cannot be identified from the data and is therefore a judgement value. The daily number of requests of each origin-destination pair is given by:

$$R_{od} = k\, f \sum_{d} \omega_d\, r_{od}(d), \tag{C.4}$$

where the sum covers the eligible departures d on valid vehicle-days and $r_{od}(d)$ is the number of requests that departure d generates on the pair. Since k is derived from the same departures, load factors and weights, the terminal requests at the hub equal the facility boardings by construction. Moreover, demand is generated only for servable pairs, i.e., pairs that lie in order on an eligible route of an agent.

The daily requests of each pair are distributed over the hours of the day according to the profile of its direction class (peak, counter-peak or no direction). In Wynberg, the profile is the load of the agents' eligible departures in each hour, including the evening trips home, without which the profile would have almost no evening peak. In Khayelitsha and Bellville, the departures of the agents are almost uniform over the day, whereas their moving time shows clear morning and evening peaks, i.e., at the peaks, the taxis move longer rather than more often. The profile of these communities is therefore the observed moving time of the agents in each hour, shifted earlier by half the median trip duration to approximate the boarding time and weighted by the mean load factor of the class in the period. Wynberg retains the departure-based timing with which it was calibrated and accepted. In the engine, the number of requests of each pair in each 15-minute interval is drawn from a Poisson distribution with uniformly distributed request times, which yields a non-homogeneous Poisson process with a piecewise-constant rate. The same profile applies to every simulated day. The resulting demand levels are given in Table 7, and the demand is audited against the corridor survey counts and the published boardings at facilities outside the communities.

Appendix C.8. Engine corrections in this study

Table C.26 lists the corrections of the charging behaviour of the engine that this study made. Each correction replaces the earlier behaviour, and no setting restores it. Since ICE drivers never charge, the corrections do not change the ICE runs: a frozen ICE run gives identical logs with the corrected engine.

Table C.26: Corrections of the charging behaviour of the engine.

Correction	**Defect that it corrects**
A driver at a station takes a connector only when its charging policy requests charging and its shift has not ended	Sessions held a connector without a need for energy, and drivers charged after their shift end
The energy of a charging step is credited before the policy decides whether to continue	Each session lost the energy of its last charging step
An EV starts a route only when its energy covers the drive to the first stop, the route, the drive from the last stop to the nearest station and the reserve; otherwise it charges first	EVs started routes that they could not finish and stranded
An EV charges before it returns home when it would otherwise arrive below the reserve	EVs stranded on the drive home
Before its shift end, a driver at a station also charges while its energy is below the need of its cheapest route	A waiting driver left the queue when its policy stopped requesting a charge, and the route test sent it back to the end of the queue
An EV does not select a route that it cannot finish from a full battery	Long routes caused strandings that no charging could prevent

Appendix D. Calibration targets and search

Appendix D.1. Targets and fit

The calibration targets are computed from the qualified vehicles on the calibration days and comprise four families: the share of agents moving in each hour of the day; the 10th, 50th and 90th percentiles of the daily distance; the same percentiles of the daily hours moving (GPS steps at 5 km/h or faster, the same threshold

that separates a stand from a hop in the travel times); and the same percentiles of the coefficient of variation of each agent's daily distance. Additionally, at least 95% of the requests must be served (a constraint), and the median seat occupancy should be as high as possible up to 0.6 (a goal), which reflects full departures from the rank and partly empty returns. The noise σ_i of each target is the standard deviation of its statistic over 500 bootstrap resamples of whole calibration dates, which preserves the dependence between vehicles on the same date. The fit of target i is given by:

$$z_i = \frac{\left| y_i^{\mathrm{sim}} - y_i^{\mathrm{obs}} \right|}{\max\left(\sigma_i,\ \epsilon_i\right)}, \tag{D.1}$$

where y_i^{sim} and y_i^{obs} are the simulated and observed values of the target, and ϵ_i is a floor (0.01 for the movement share, 5 km, 0.02 for the coefficient of variation and 0.25 h) that prevents targets with very small noise from dominating the fit. The loss of a family is the mean fit of its targets, and the loss of a run is the weighted mean of the family losses.

Appendix D.2. Search

Following [47], the calibration proceeds in two stages. First, hand tuning from the warm start locates a starting point within the region of plausible operation and identifies gaps that no behavioural parameter can close. Such gaps required changes to model rules, namely the travel-time scaling and the last-run rule in Khayelitsha. Second, a search strategy is frozen in a versioned file whose SHA-256 digest is recorded. The strategy specifies nine parameters and their bounds (the idle bias and idle demand weight; the stop bias and maximum stop duration; the first-stop bias, demand weight and maximum duration; and the shift spread and shift end offset), the objectives, the constraint and the budget. The Khayelitsha strategy weights the movement family by 0.25 relative to the duty-cycle families, since a search with equal weights traded the duty cycles for the hourly shape of the day, whereas the Bellville and Wynberg strategies use equal weights and apply this priority at the acceptance. The search uses the multivariate tree-structured Parzen estimator (TPE) of Optuna [80] with two objectives, namely the minimisation of the loss and the maximisation of the occupancy capped at 0.6, subject to the served-share constraint. Each trial runs one seed from the warm start, and the search starts with a batch of 40 trials, of which 10 are space-filling start trials, followed by optional continuation batches of 20 trials. Selected points on the feasible front are subsequently validated with cold runs, which include the full warm-up, on three fresh seeds.

Table D.27 gives the result of the search in each community. The search loss is that of one seed from the warm start; the cold loss is the mean of three cold runs. A lower search loss does not always give a lower cold loss. In Khayelitsha, the search started from the calibrated values of Wynberg. The best search trial had a lower search loss than the start point (1.785 against 1.887), but a higher cold loss (2.010 against 1.988), so the start point was accepted. In Wynberg, the accepted trial has a higher search loss than the start point but a lower cold loss.

Table D.27: Calibration search of each community: the start point, the search trial with the lowest search loss and the accepted point. Trial 0 is the start point. Trials: the completed search trials. Search loss: one seed from the warm start. Cold loss: the mean of three cold runs with the full warm-up. A dash shows a point that was not validated with cold runs.

Community	Point	Trial	Trials	Search loss	Cold loss
Khayelitsha	Start	0	40	1.887	1.988
	Best search trial	17		1.785	2.010
	Accepted	0		1.887	1.988
Bellville	Start	0	32	2.326	–
	Best search trial	20		1.901	2.067
	Accepted	28		1.914	2.002
Wynberg	Start	0	39	1.894	2.476
	Best search trial	0		1.894	2.476
	Accepted	25		1.985	2.377

Appendix D.3. Assessment and hold-out

Each validated point is assessed with graded levels (good, acceptable or limited), which are expressed relative to the loss of the qualified vehicles against their own targets over resampled calibration dates, so that the losses of the three communities are comparable. The acceptance prioritises realistic behaviour and the duty-cycle families, followed by the stability across seeds, the served share and the occupancy, and finally the hourly shape of the day. After the acceptance, a hold-out report compares the fit of the accepted model against the targets of the hold-out days (2 and 10 December) with its fit against the calibration targets. The resulting hold-out gap is assessed against the loss of the observed hold-out days against the calibration targets, which quantifies how much two days differ from six days in the data alone. The hold-out days test only the calibrated behaviour, since the network, the communities and the routes use all study weekdays.

Appendix E. Upper bound of the design space

The design space allows up to $2\hat{N}_j$ connectors at hub j (Section 4.7.1). In Khayelitsha, the best validated design, 19-8-6, reaches this bound at hubs 2 and 3. Three designs above the bound, up to $3\hat{N}_j$, are therefore evaluated in M_A on the five selection seeds (Table E.28). None of them improves the EV passenger change: the differences to the best design are -0.1 to -1.0 percentage points, within two standard errors. The designs 19-12-9 and 25-12-9 give identical runs, because the drivers do not use the additional connectors at hub 1. In M_R, all four designs strand the same share of the vehicle-days. The bound therefore does not constrain the selection.

Table E.28: The best validated Khayelitsha design and three designs above the bound $2\hat{N}_j = (20, 8, 6)$, with 50 kW connectors. M_A: the means over the five selection seeds, and the paired difference to the best design with its standard error. M_R: the stranded share of the vehicle-days and the peak site load.

		M_A			M_R	
Design	Installed (kW)	Peak (kW)	EV passenger change (SE)	Difference to best (SE)	Stranded (share)	Peak (kW)
19-8-6 (best)	1,650	860	−0.027 (0.009)	–	0.549	500
19-10-8	1,850	910	−0.037 (0.007)	−0.010 (0.007)	0.549	500
19-12-9	2,000	890	−0.029 (0.009)	−0.001 (0.003)	0.549	500
25-12-9	2,300	890	−0.029 (0.009)	−0.001 (0.003)	0.549	500

Appendix F. Reproducibility

The study code is in the `emanalysis` repository, in the project `embt_replay_vs_abm`. The router telemetry is private and is not published; the paper reports aggregates only. Each step of the pipeline (`src/pipeline.py`) writes its outputs with a manifest that records the digests of its inputs, its settings, its code and its outputs, so a changed input marks every later step as out of date. Each step also has validity checks and a notebook in which the researcher records the decision of the step, for example the network review, the community partition and the accepted calibration; the selection of the charging designs follows the rules of Section 4.7.2 with no manual choice.

The agent-based model is the engine `parasim-engine`, branch `charging-corrections`, commit `fd5e946`. The SHA-256 of the engine executable in the manifests of all experiment runs is

`44bb30f2b511e3ea70152a710a7b1bb5ecd2f4fb8324a7ecab0a9bc8f2652eda`.

The engine seed of a run is 101 plus its seed offset: the design search uses seed 101, the selection uses the seeds 102 to 106, the cross-evaluation uses the fresh seeds 201 to 210 and the sensitivity cases use 201 to 205.